

Model Analysis

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1 Data analysis /Dataframe import and filters

1.1 DataFrame import and filters

The data has been imported with the following(s) function(s):

```
def build_df(data):
    if flag_IR:
        df, features_engineered = import_data_IR()
        if flag_IR_maskP:
            df = mask_period_IR(df)
        if flag_IR_balR:
            df = mask_balanceHC_random_IR(df, n_samples=None)
    if flag_ER:
        df, features_engineered = import_data_ER()

    # Removing of the columns where there is more than 15% of nan
    #df_features = pd.DataFrame(features_all)
    cols_to_check = target_col + features_X
```

```

nan_ratio = df[cols_to_check].isna().mean()
valid_features = nan_ratio[nan_ratio <= 0.15].index
valid_features_list = list(valid_features)

#df = df[valid_features]
# Removing of last np.nan
mask = np.isfinite(df[valid_features]).all(axis=1)
df = df[mask]

cols = list(dict.fromkeys(valid_features_list + #target_col + features_X +
                        feats_to_balance + feats_grouping +
                        feats_reference+ feats_postproc))

df = df[cols]
#df['campaign'] = "ALL"

new_feats_X = [x for x in features_X if x in valid_features_list]
new_feats_add = [x for x in feats_eng if x in valid_features_list]

data["df"]=df
data["features_X"] = new_feats_X
data["feats_added"] = new_feats_add

return data

```

```

def import_data_IR():
    if target_col[0] == "i_NH3":
        file_path = "csv_decomposed/innova/df_signal_decomposition_NH3_junesept.csv"
        #file_path = "csv/innova/switched/df_signal_decomposition_10_50_200.csv"
    elif target_col[0] == "i_CO2":
        file_path = "csv_decomposed/innova/df_signal_decomposition_CO2_current.csv"
    else:
        raise ValueError("Target is not supported")

    df = pd.read_csv(file_path, sep=";")
    df["datetime"] = pd.to_datetime(df["datetime"])
    df.sort_values("datetime").reset_index(drop=True)

    if target_col[0] == "i_NH3":
        df["campaign"] = df["datetime"].apply(assign_campaign)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point, axis=1)

```

```

df = df[df["innova_point"] != "no"].copy()
with open("csv_decomposed/innova/features_decomposition_NH3_list_junesept.json", "r") as f:
    #with open("csv/innova/switched/features_decomposition_list_10_50_200.json", "r") as f:
        dict_features = json.load(f)
elif target_col[0] == "i_CO2":
    df["campaign"] = df["datetime"].apply(assign_campaign_iCO2)
    df = df[df["campaign"] != "other"].copy()
    df["innova_point"] = df.apply(assign_innova_point_iCO2, axis=1)
    df = df[df["innova_point"] != "no"].copy()
    with open("csv/innova/features_decomposition_CO2_list.json", "r") as f:
        dict_features = json.load(f)
else:
    raise ValueError("The target col do not match any existing function")

df["hot"] = df.apply(assign_hot_season,axis=1)
df["umidity_range"] = df.apply(assign_humidity, axis = 1)
#Removing of rabbit 4
df["column"] = df.apply(assign_colonna, axis = 1)
df['height'] = df.apply(assign_altezza, axis = 1)

return df,dict_features

```

```

def assign_innova_point(row):
    camp = row["campaign"]
    rid = row["id_rabbit"]

    # if camp == "2025_Jun":
    #     mapping = {2: "sl11", 4: "sl5", 6: "sl4", 1: "sl3", 5: "no"}
    # elif camp == "2025_Sep":
    #     mapping = {2: "sl11", 4: "sl5", 3: "sl4", 6: "sl3", 5: "no"}
    if camp == "2025_Apr":
        #mapping = {1: "no", 2: "no", 4: "no", 6: "sl5"}
        #mapping = {1: "no", 2: "sl5", 4: "no", 6: "sl5"}
        mapping = {1: "no", 2: "no", 4: "no", 6: "no"}
    elif camp == "2025_Jun":
        mapping = {2: "sl11", 4: "sl5", 1: "sl3", 6: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 1: "sl4", 6: "sl3", 5: "no"}
    elif camp == "2025_Sep":
        mapping = {2: "sl11", 4: "sl5", 6: "sl3", 3: "sl4", 5: "no"}

```

```

        #mapping = {2: "sl11", 4: "no", 6: "sl4", 3: "sl3", 5: "no"}
    elif camp == "2026_Gen":
        #mapping = {1: "no", 4: "no", 5: "sl11", 6: "no"}
        #mapping = {1: "sl11", 4: "no", 5: "sl11", 6: "sl11"}
        mapping = {1: "no", 4: "no", 5: "no", 6: "no"}
    else:
        mapping = {}

    return mapping.get(rid, "no")

```

1.2 Metadata

Nombre de lignes df	8033
Nombre de lignes df clean	8033
Nombre de features	20
% de NaN	0.0
Taille dataset entraînement	5623
Taille dataset test	2409
Métrique utilisée	r2

Column target : i_NH3

Features_X : r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_wl_2h_D1_energy, r_NH3_wl_2h_A_energy, r_NH3_wl_2h_A_rms, r_NH3_wl_2h_D1_energy_rel, r_NH3_wl_2h_A_energy_rel, r_temperature_wl_2h_D1_energy, r_temperature_wl_2h_A_energy, r_temperature_wl_2h_A_rms, r_temperature_wl_2h_D1_energy_rel, r_temperature_wl_2h_A_energy_rel, r_umidity_wl_2h_D1_energy, r_umidity_wl_2h_A_energy, r_umidity_wl_2h_A_rms, r_umidity_wl_2h_D1_energy_rel, r_umidity_wl_2h_A_energy_rel

feats_to_balance :

feats_grouping :

feats_postproc : datetime, id_channel, id_rabbit, campaign

feats_ref :

id_cols : campaign

2 Data correlation and clustering

2.1 Clustermap and cluster list

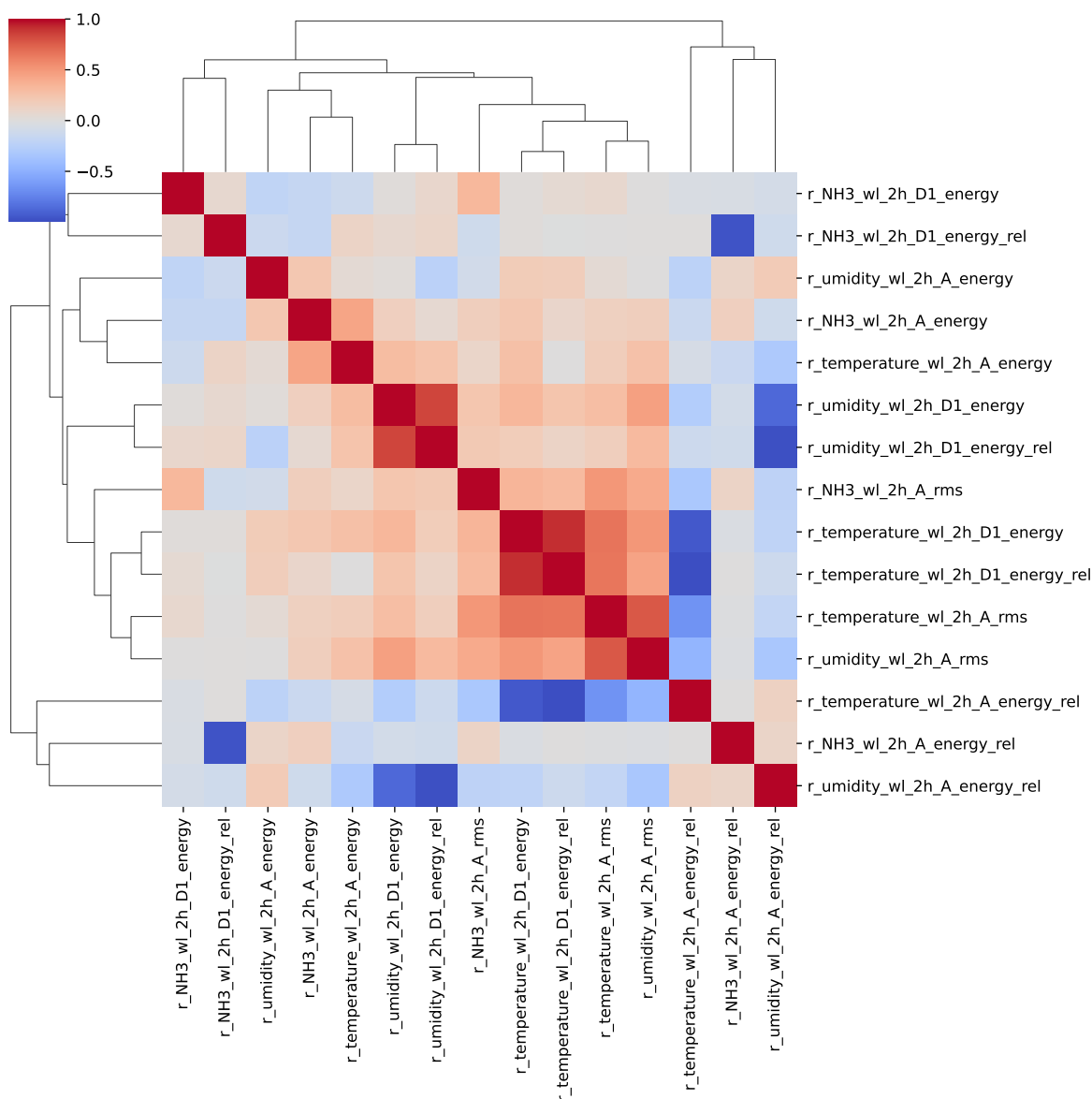
The following section has been done with a threshold choosen of : 0.8

cluster 1: r, , N, H, 3 ; **cluster 2:** r, , t, e, m, p, e, r, a, t, u, r, e ; **cluster 3:** r, , u, m, i, d, i, t, y ; **cluster 4:** r, , C, O, 2 ; **cluster 5:** r, __, T, H, I ; **cluster 6:** r_NH3_wl_2h_D1_energy ; **cluster 7:** r_NH3_wl_2h_A_energy ; **cluster 8:** r_NH3_wl_2h_A_rms ; **cluster 9:** r_NH3_wl_2h_A_energy_rel, r_NH3_wl_2h_D1_energy_rel ; **cluster 10:** r_temperature_wl_2h_A_energy_rel, r_temperature_wl_2h_D1_energy_rel ; **cluster 11:** r_temperature_wl_2h_A_energy ; **cluster 12:** r_temperature_wl_2h_A_rms ; **cluster 13:** r_umidity_wl_2h_D1_energy_rel, r_umidity_wl_2h_D1_energy, r_umidity_wl_2h_A_energy_rel ; **cluster 14:** r_umidity_wl_2h_A_energy ; **cluster 15:** r_umidity_wl_2h_A_rms ;

2.2 Features selection

Only the first items of the clusters is choosen for the rest of the study

Choosen features_X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_wl_2h_D1_energy, r_NH3_wl_2h_A_energy, r_NH3_wl_2h_A_rms, r_NH3_wl_2h_A_energy_rel, r_temperature_wl_2h_A_energy_rel, r_temperature_wl_2h_A_energy, r_temperature_wl_2h_A_rms, r_umidity_wl_2h_D1_energy_rel, r_umidity_wl_2h_A_energy, r_umidity_wl_2h_A_rms



3 Model(s) analysis

3.1 Metadata

Features_X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_wl_2h_D1_energy, r_NH3_wl_2h_A_energy, r_NH3_wl_2h_A_rms, r_NH3_wl_2h_A_energy_rel,

r_temperature_wl_2h_A_energy_rel, r_temperature_wl_2h_A_energy, r_temperature_wl_2h_A_rms, r_umidity_wl_2h_D1_energy_rel, r_umidity_wl_2h_A_energy, r_umidity_wl_2h_A_rms

Len(Features_X): 15

Config Gridsearch:{‘show’: True, ‘search_type’: ‘optuna’, ‘model_name’: None, ‘model’: None, ‘param_grid’: None, ‘best_param’: None, ‘cv_func’: ShuffleSplit(n_splits=5, random_state=42, test_size=0.3, train_size=None), ‘save_models’: True, ‘sumup-table’: True}

3.2 Explanation legend tables

Columns

- rank: ranking regarding cross validation test
- train_cv: mean score train of the CV
- test_cv: mean score test of the CV
- test: score on the testing dataset
- diff: test-test_cv
- r_test: test/test_cv
- r_train: train_cv/test_cv

Colors:

[Orange] $r_{\text{test}} < 0.95$ (under performance of the test)

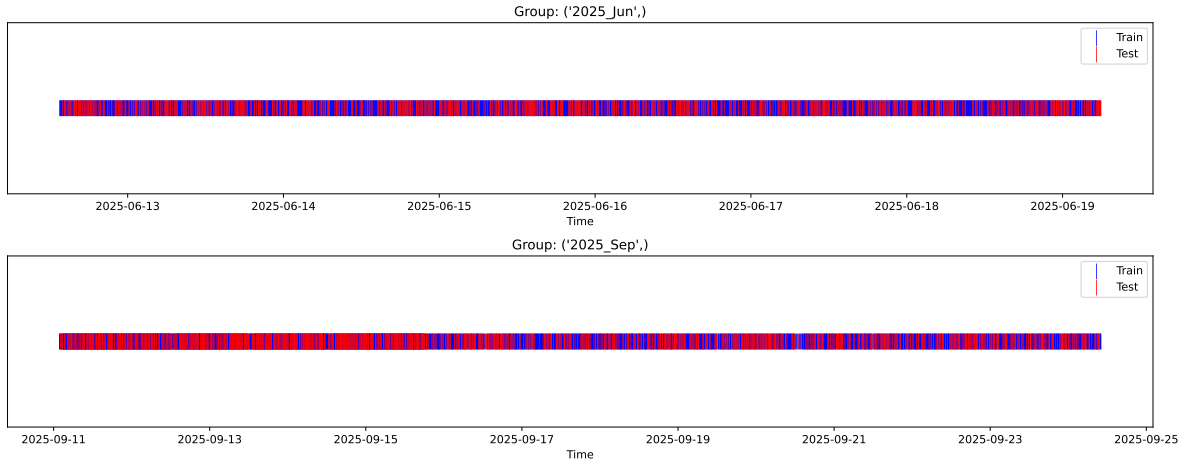
[Green] $r_{\text{test}} > 1.05$ (over performance of the test)

[Red] r_{train} not in $[0.95, 1.05]$ (gap between train/test)

3.3 Repartition of train and test

3.3.1 External Split: X_train vs X_test

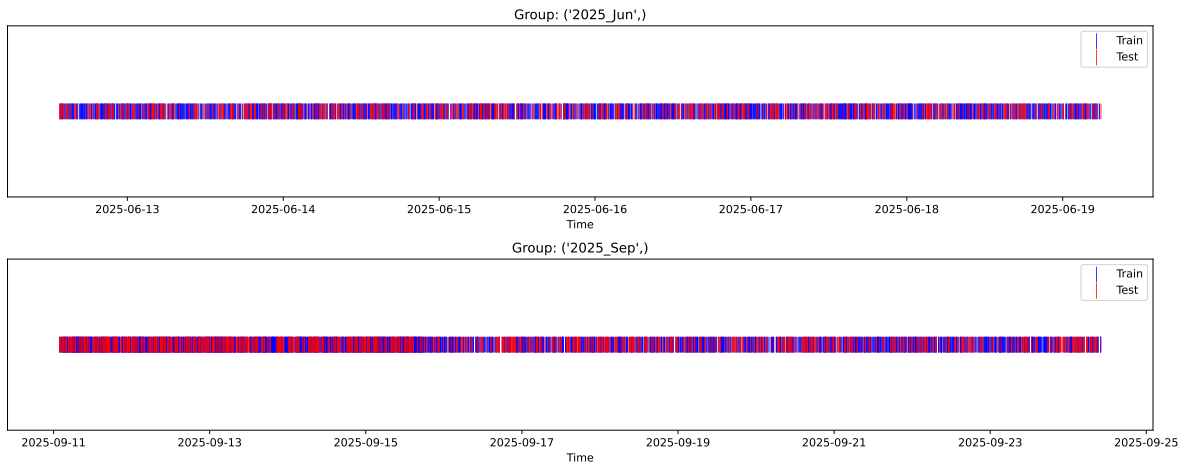
External Split: Train: 5623 samples Test: 2410 samples



3.3.2 Internal CV Folds

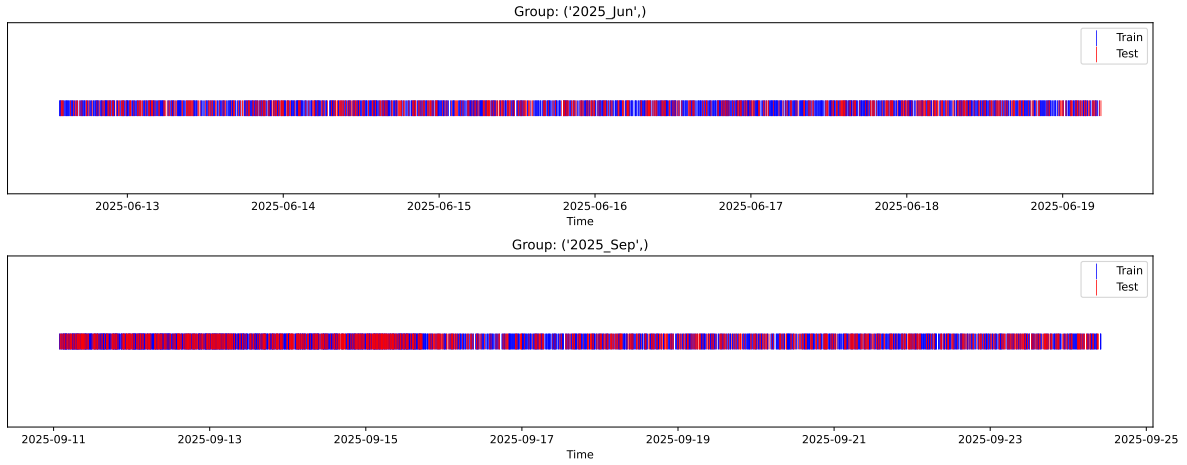
Fold 1/5

Train: 3936 samples Test: 1687 samples Excluded: 0 samples Window exclusion: 0h



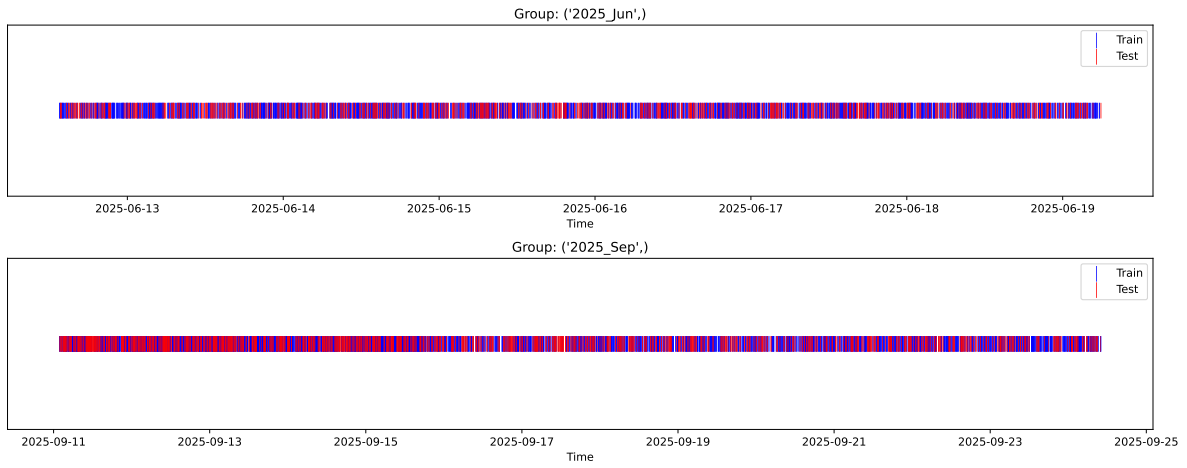
Fold 2/5

Train: 3936 samples Test: 1687 samples Excluded: 0 samples Window exclusion: 0h



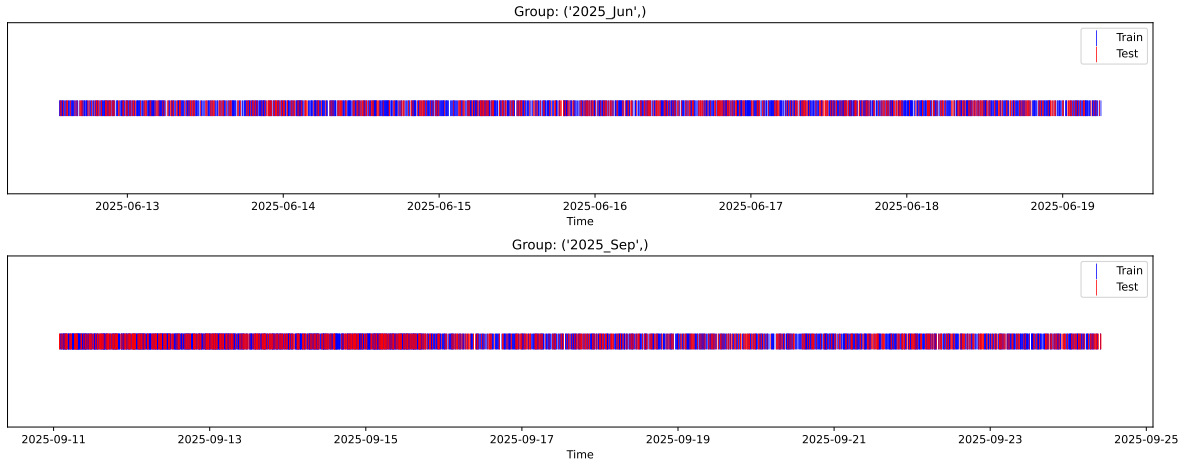
Fold 3/5

Train: 3936 samples Test: 1687 samples Excluded: 0 samples Window exclusion: 0h



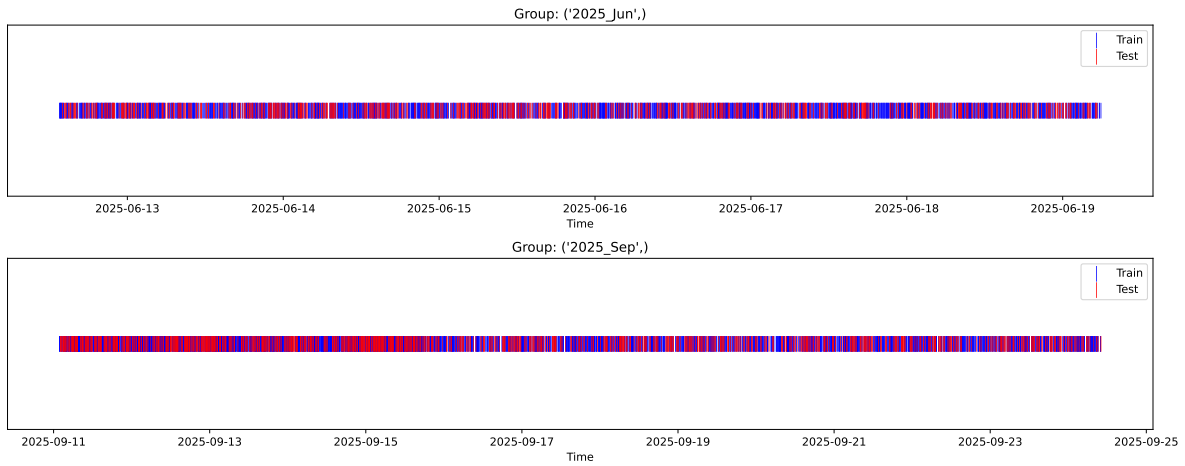
Fold 4/5

Train: 3936 samples Test: 1687 samples Excluded: 0 samples Window exclusion: 0h



Fold 5/5

Train: 3936 samples Test: 1687 samples Excluded: 0 samples Window exclusion: 0h



3.4 Model : Linear

3.4.1 Gridsearch

Distribution y_{train} vs y_{test} : y_{train} : mean=2.345, std=0.801, min=0.959, max=13.845
 y_{test} : mean=2.349, std=0.784, min=1.071, max=7.822

Fold 0 — y_{test} : mean=2.34, std=0.78, y_{train} : mean=2.35, std=0.81

Fold 1 — y_{test} : mean=2.34, std=0.78, y_{train} : mean=2.35, std=0.81

Fold 2 — y_{test} : mean=2.37, std=0.85, y_{train} : mean=2.34, std=0.78

Fold 3 — y_test: mean=2.34, std=0.78, y_train: mean=2.35, std=0.81

Fold 4 — y_test: mean=2.35, std=0.77, y_train: mean=2.34, std=0.81

train scores CV: [0.50242618 0.51915841 0.52857687 0.51474094 0.51890772]

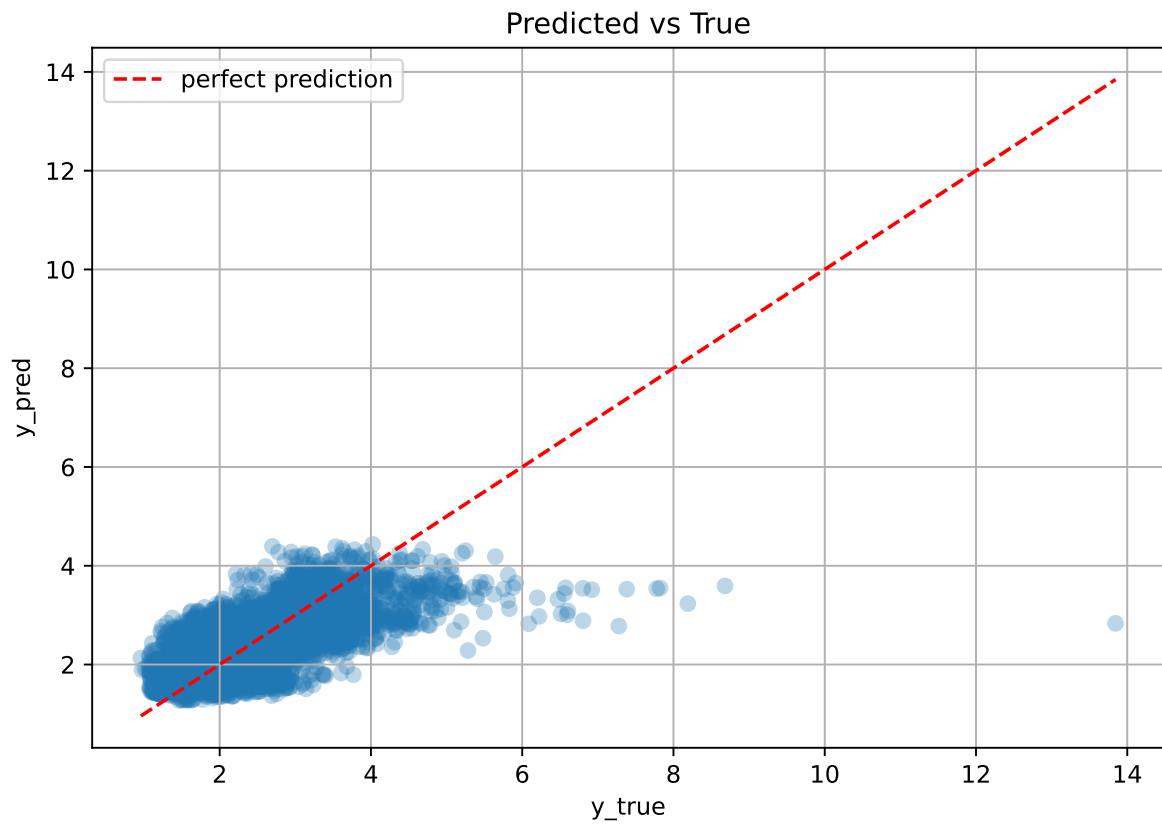
test scores CV: [0.56959883 0.52892036 0.50832721 0.53824755 0.52796519]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.5168	0.5346	0.02	0.4947	0.4954	-0.0392	0.9267	0.9666

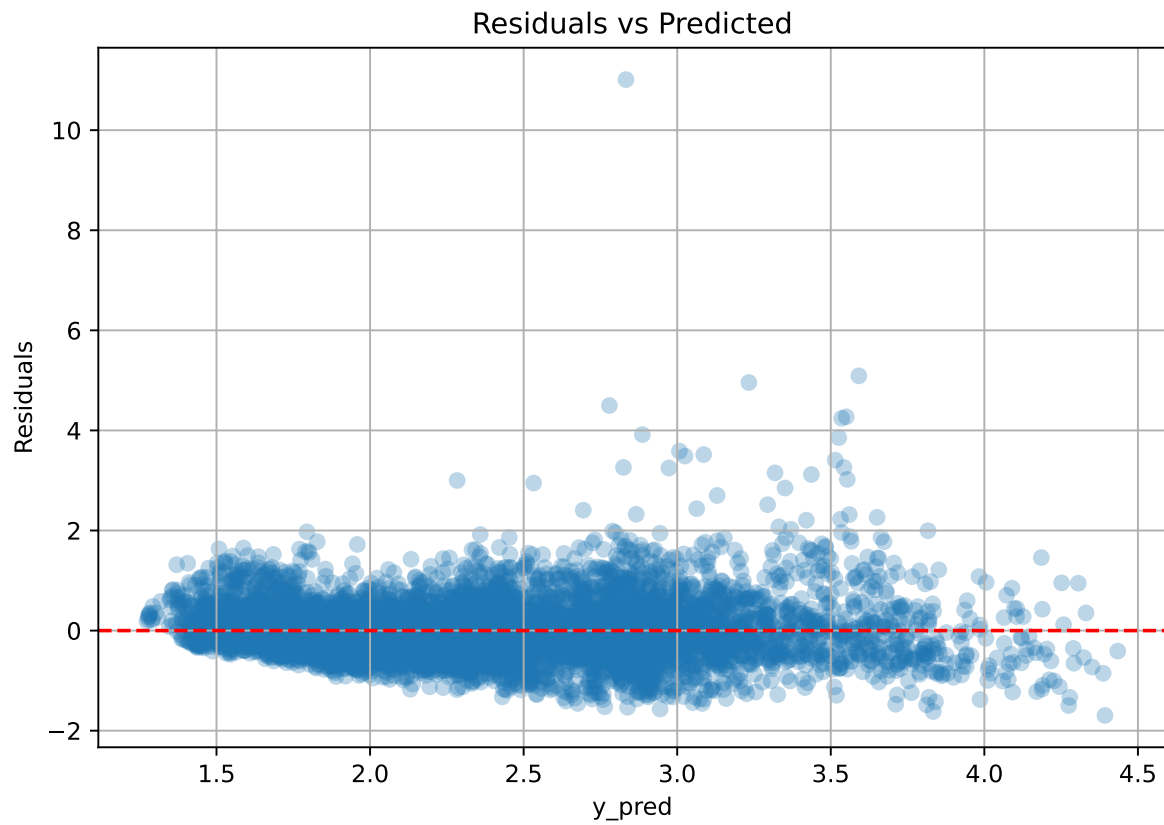
Coefficients:

coef_r_CO2 = -0.0465, coef_r_NH3 = 0.1322, coef_r_NH3_wl_2h_A_energy = -0.2291, coef_r_NH3_wl_2h_A_energy_rel = 0.0045, coef_r_NH3_wl_2h_A_rms = 0.0247, coef_r_NH3_wl_2h_D1_energy = 0.0160, coef_r_THI = 0.4042, coef_r_temperature = -0.3285, coef_r_temperature_wl_2h_A_energy = -0.1517, coef_r_temperature_wl_2h_A_energy_rel = -0.0644, coef_r_temperature_wl_2h_A_rms = 0.1518, coef_r_umidity = -0.6257, coef_r_umidity_wl_2h_A_energy = 0.2428, coef_r_umidity_wl_2h_A_rms = 0.0524, coef_r_umidity_wl_2h_D1_energy_rel = 0.0324, intercept = 2.3453,

3.4.2 Scatter



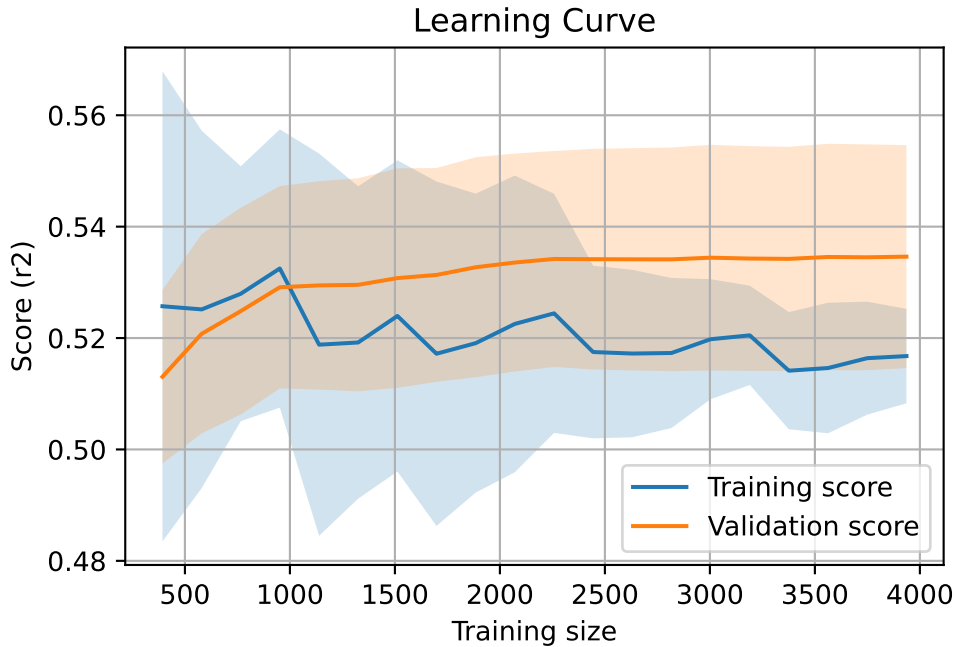
3.4.3 Scatter residuals



3.4.4 Learning curve

`Len(group)` : 8033, `Len(training)` : 5624,

`Len(training_real)` : 3937, `Len(test_of_training)` : 1687, `Len(test)` : 2409



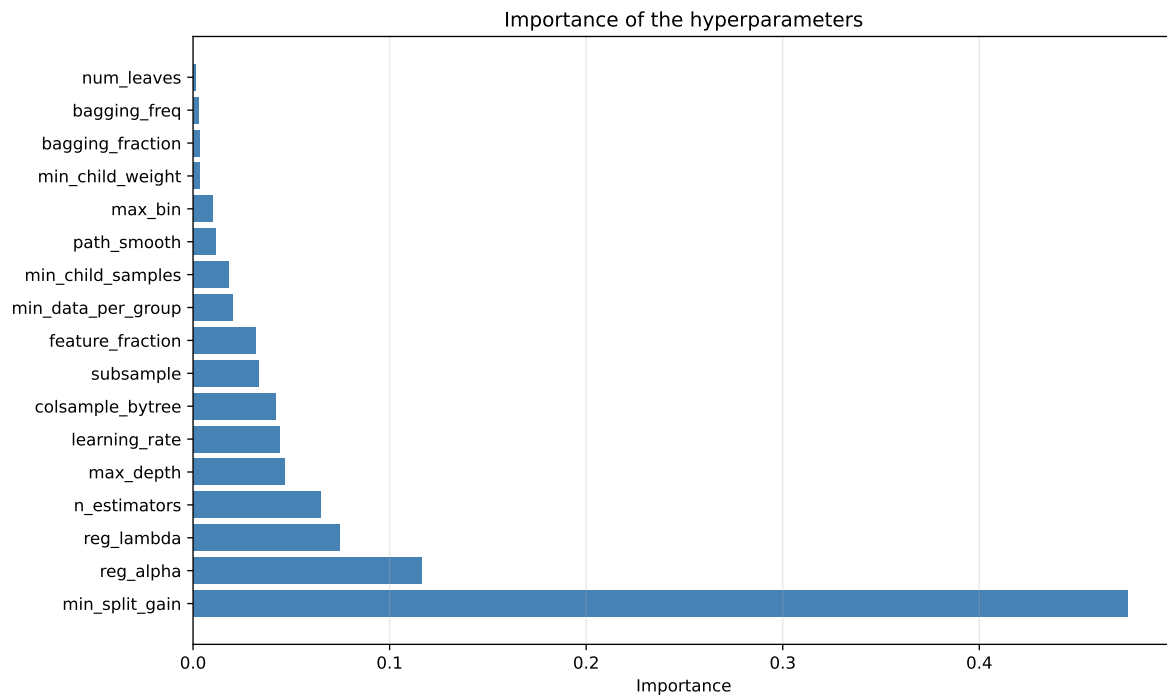
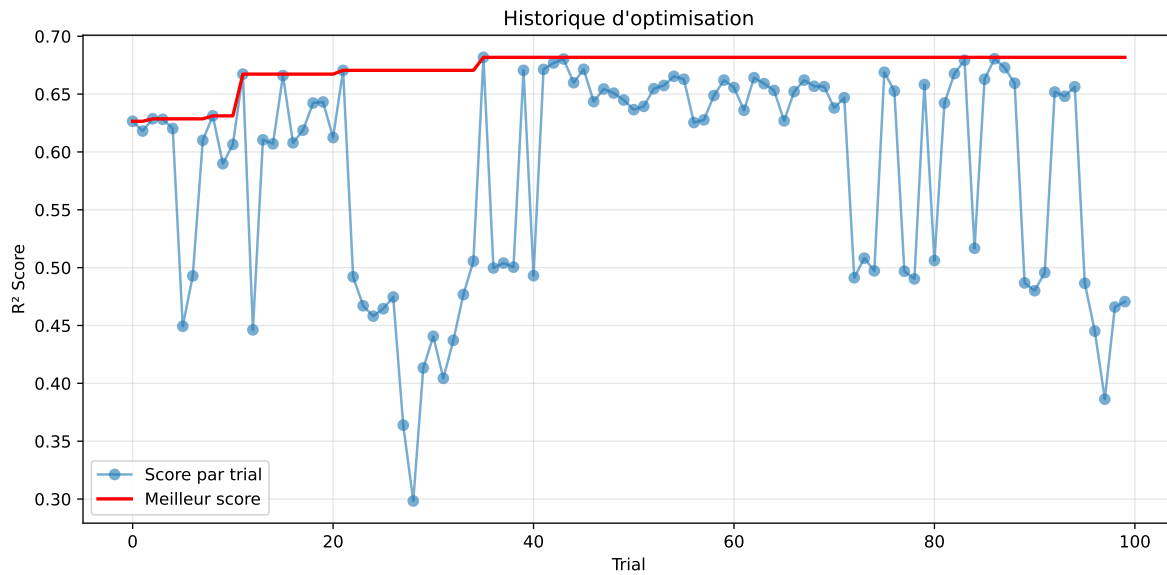
3.5 Model : LGBM

3.5.1 Gridsearch

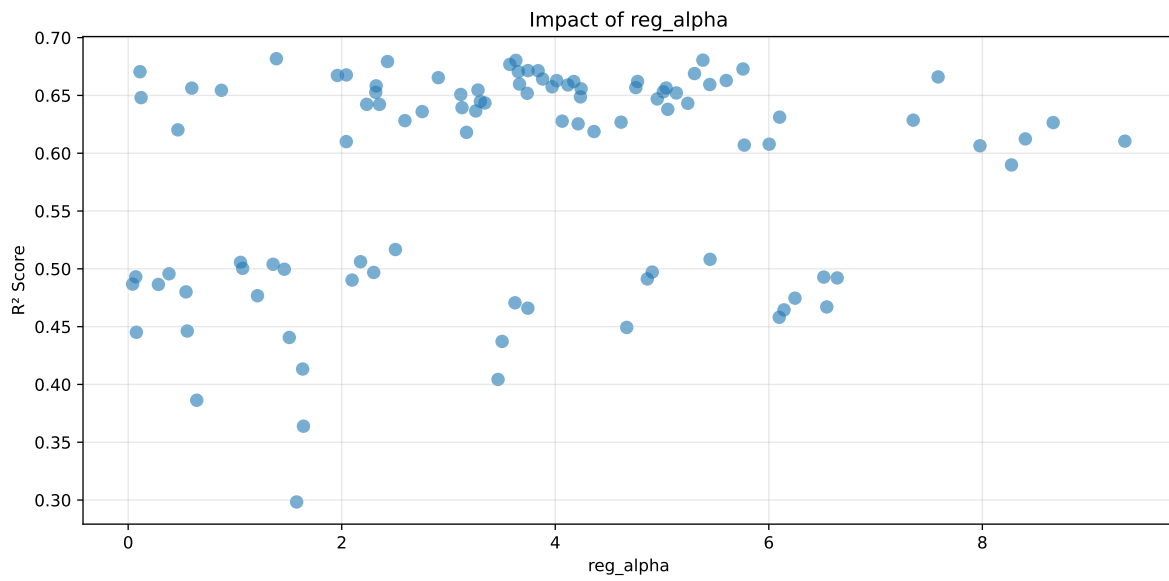
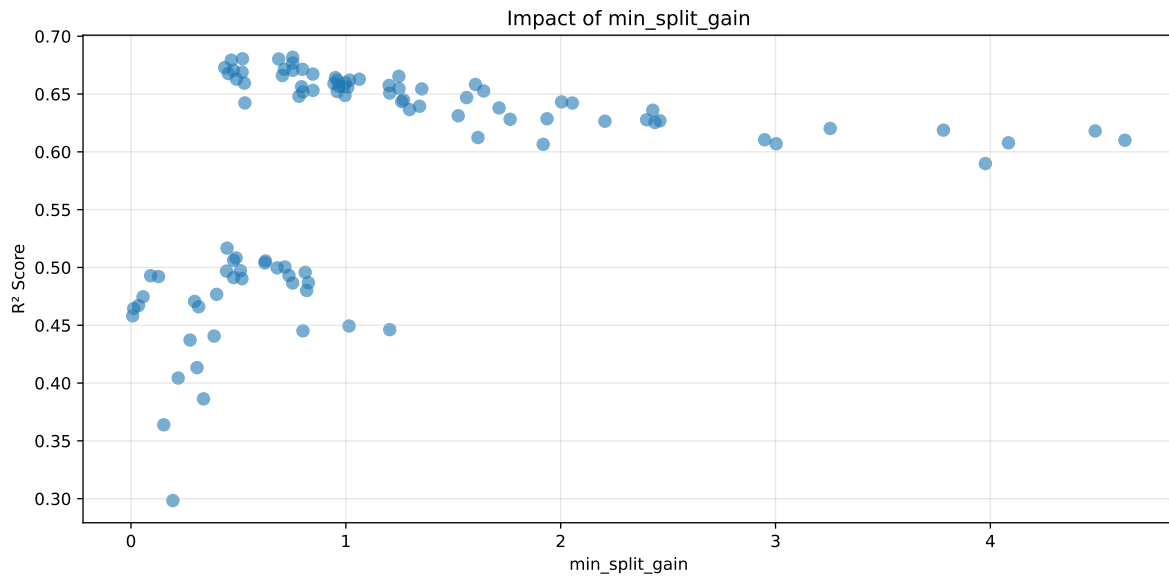
Distribution y_train vs y_test: y_train: mean=2.345, std=0.801, min=0.959, max=13.845
y_test: mean=2.349, std=0.784, min=1.071, max=7.822

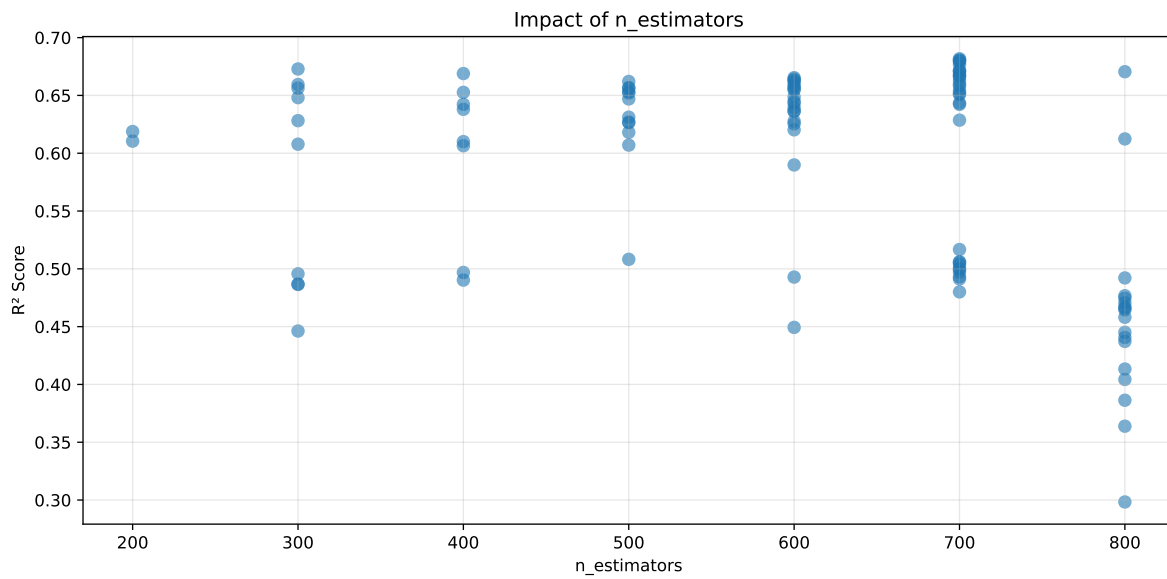
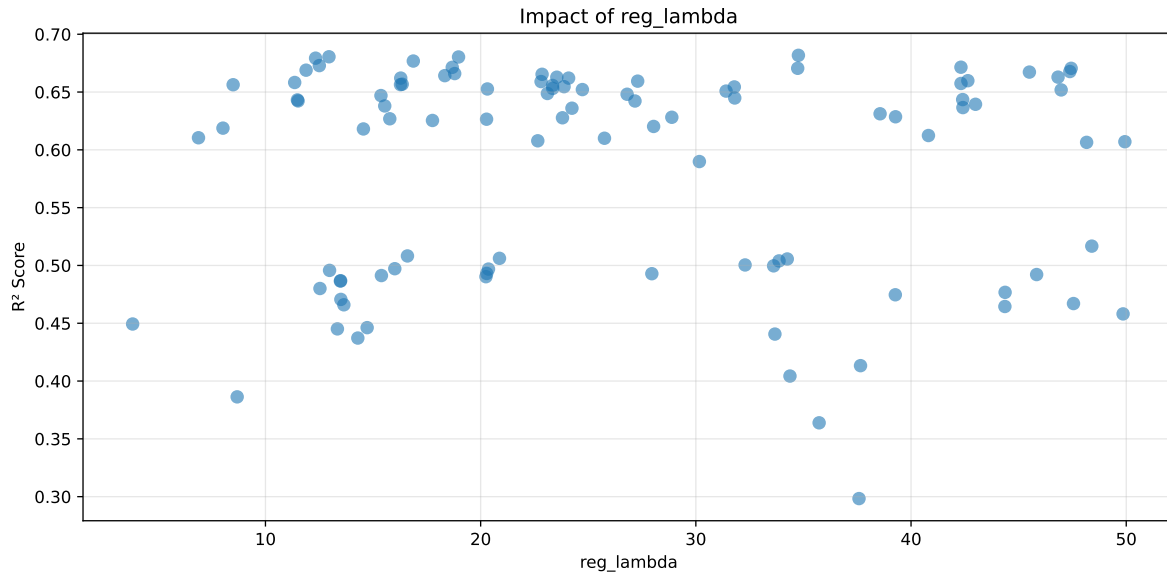
===== TOP Importance FEATURES =====

feature importance r_CO2 1084 r_NH3_wl_2h_A_energy 698 r_NH3 670 r_umidity 610
r_temperature_wl_2h_A_energy 559 r_temperature_wl_2h_A_rms 458 r_temperature
441 r_umidity_wl_2h_A_energy 411 r_THI 366 r_temperature_wl_2h_A_energy_rel 352
r_NH3_wl_2h_D1_energy 294 r_NH3_wl_2h_A_rms 265 r_umidity_wl_2h_A_rms 239
r_umidity_wl_2h_D1_energy_rel 193 r_NH3_wl_2h_A_energy_rel 178



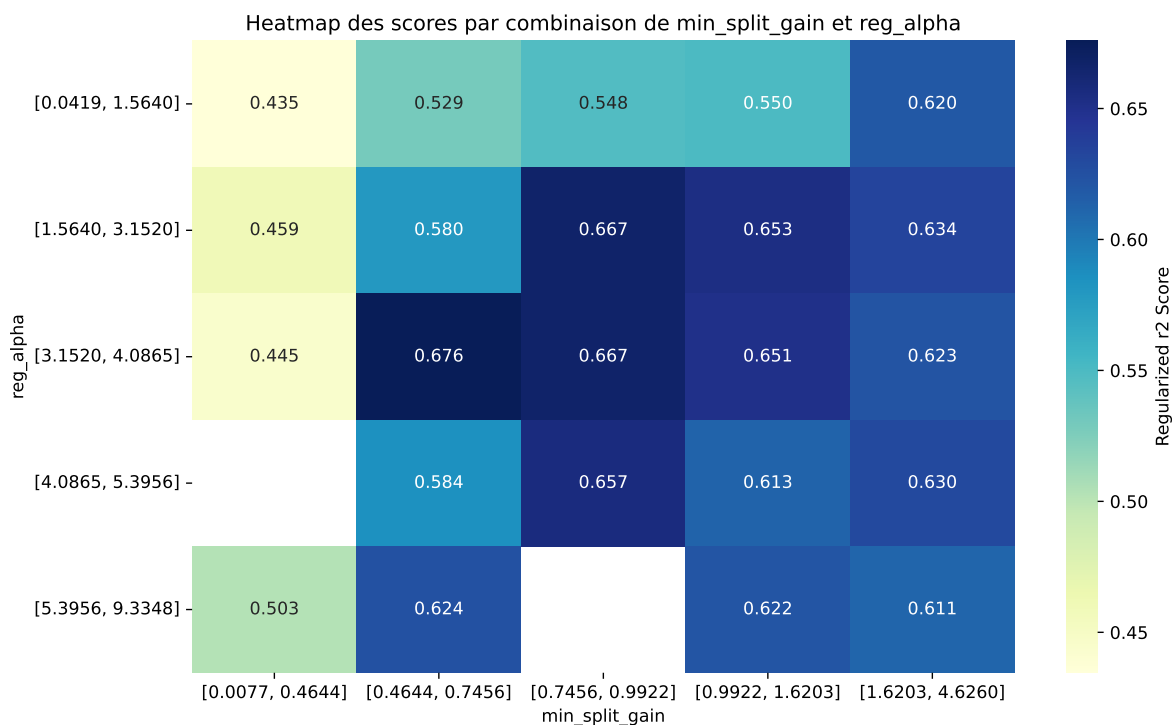
Paramètres avec >5% d'importance : ['min_split_gain', 'reg_alpha', 'reg_lambda', 'n_estimators']





C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.7118	0.6818	0.6514	0.0238	0.6818	0.6721	-0.0097	0.9858	1.0441
2	0.7102	0.6805	0.6518	0.0232	0.6805	0.6716	-0.0089	0.9869	1.0437
3	0.7118	0.6803	0.6509	0.0227	0.6803	0.6692	-0.0111	0.9836	1.0463
4	0.7085	0.6793	0.6478	0.0231	0.6793	0.6541	-0.0252	0.9629	1.043
5	0.7061	0.6768	0.6468	0.024	0.6768	0.6634	-0.0134	0.9802	1.0433
6	0.7032	0.6729	0.6424	0.0239	0.6729	0.6625	-0.0104	0.9846	1.045
7	0.6962	0.6715	0.6411	0.0254	0.6715	0.6622	-0.0093	0.9862	1.0369
8	0.6998	0.6714	0.6416	0.0214	0.6714	0.6632	-0.0081	0.9879	1.0424
9	0.6967	0.6705	0.6403	0.0248	0.6705	0.6571	-0.0134	0.98	1.039
10	0.6947	0.6705	0.6396	0.0232	0.6705	0.6511	-0.0195	0.971	1.0361
---	---	---	---	---	---	---	---	---	---
100	0.8312	0.7424	0.7089	0.0269	0.2983	0.7297	-0.0127	0.9829	1.1196

Hyperparameters:

Rank 1: {'reg__n_estimators': 700, 'reg__max_depth': 8, 'reg__learning_rate': 0.014558397919091389, 'reg__num_leaves': 25, 'reg__subsample': 0.6271269725784868, 'reg__colsample_bytree': 0.7322377051064219, 'reg__min_child_samples': 27, 'reg__reg_alpha': 1.3882438294733799, 'reg__reg_lambda': 34.76590655698346, 'reg__min_split_gain': 0.7526702957194722, 'reg__path_smooth': 2.008694510162571, 'reg__min_child_weight':

0.08620584920632732, 'reg__feature_fraction': 0.5156434794030417, 'reg__bagging_fraction': 0.5264257334474702, 'reg__bagging_freq': 2, 'reg__max_bin': 93, 'reg__min_data_per_group': 162}

Rank 2: {'reg__n_estimators': 700, 'reg__max_depth': 7, 'reg__learning_rate': 0.017505418734197305, 'reg__num_leaves': 17, 'reg__subsample': 0.6587367444706532, 'reg__colsample_bytree': 0.6896392471761946, 'reg__min_child_samples': 29, 'reg__reg_alpha': 5.38268645432462, 'reg__reg_lambda': 12.951754020651425, 'reg__min_split_gain': 0.5190648204616336, 'reg__path_smooth': 4.409609042621884, 'reg__min_child_weight': 0.1010671562537856, 'reg__feature_fraction': 0.6490906350604834, 'reg__bagging_fraction': 0.5717995893796368, 'reg__bagging_freq': 4, 'reg__max_bin': 67, 'reg__min_data_per_group': 193}

Rank 3: {'reg__n_estimators': 700, 'reg__max_depth': 4, 'reg__learning_rate': 0.014666971953509092, 'reg__num_leaves': 15, 'reg__subsample': 0.6241593201577139, 'reg__colsample_bytree': 0.6491034539558348, 'reg__min_child_samples': 27, 'reg__reg_alpha': 3.6310565826129335, 'reg__reg_lambda': 18.972777768005702, 'reg__min_split_gain': 0.6876039959426872, 'reg__path_smooth': 1.2394194914032906, 'reg__min_child_weight': 0.0710988406878563, 'reg__feature_fraction': 0.7390364399150193, 'reg__bagging_fraction': 0.6815775298770401, 'reg__bagging_freq': 2, 'reg__max_bin': 95, 'reg__min_data_per_group': 196}

Rank 4: {'reg__n_estimators': 700, 'reg__max_depth': 3, 'reg__learning_rate': 0.012105679724323642, 'reg__num_leaves': 39, 'reg__subsample': 0.716552326665805, 'reg__colsample_bytree': 0.5820484174147742, 'reg__min_child_samples': 19, 'reg__reg_alpha': 2.4291281003574174, 'reg__reg_lambda': 12.329717765441986, 'reg__min_split_gain': 0.467153484433389, 'reg__path_smooth': 4.025467022513181, 'reg__min_child_weight': 0.00667276893583785, 'reg__feature_fraction': 0.6567290547789535, 'reg__bagging_fraction': 0.7233184587040903, 'reg__bagging_freq': 3, 'reg__max_bin': 86, 'reg__min_data_per_group': 104}

Rank 5: {'reg__n_estimators': 700, 'reg__max_depth': 4, 'reg__learning_rate': 0.014406482144318692, 'reg__num_leaves': 43, 'reg__subsample': 0.6248341460345825, 'reg__colsample_bytree': 0.5215892213430805, 'reg__min_child_samples': 24, 'reg__reg_alpha': 3.5742254178419155, 'reg__reg_lambda': 16.870636292177387, 'reg__min_split_gain': 0.7525892132700317, 'reg__path_smooth': 4.1769234019149915, 'reg__min_child_weight': 0.09189263459208383, 'reg__feature_fraction': 0.7373830814122472, 'reg__bagging_fraction': 0.6636533353175313, 'reg__bagging_freq': 2, 'reg__max_bin': 94, 'reg__min_data_per_group': 186}

Rank 6: {'reg__n_estimators': 300, 'reg__max_depth': 7, 'reg__learning_rate': 0.017247856454746035, 'reg__num_leaves': 17, 'reg__subsample': 0.5490654620156077, 'reg__colsample_bytree': 0.6892881700732283, 'reg__min_child_samples': 29, 'reg__reg_alpha': 5.758129963508731, 'reg__reg_lambda': 12.50306336996846, 'reg__min_split_gain': 0.4364472542700915, 'reg__path_smooth': 4.682056608831739, 'reg__min_child_weight':

0.0982958057183612, 'reg__feature_fraction': 0.6528024384236072, 'reg__bagging_fraction': 0.5796926630949315, 'reg__bagging_freq': 4, 'reg__max_bin': 253, 'reg__min_data_per_group': 120}

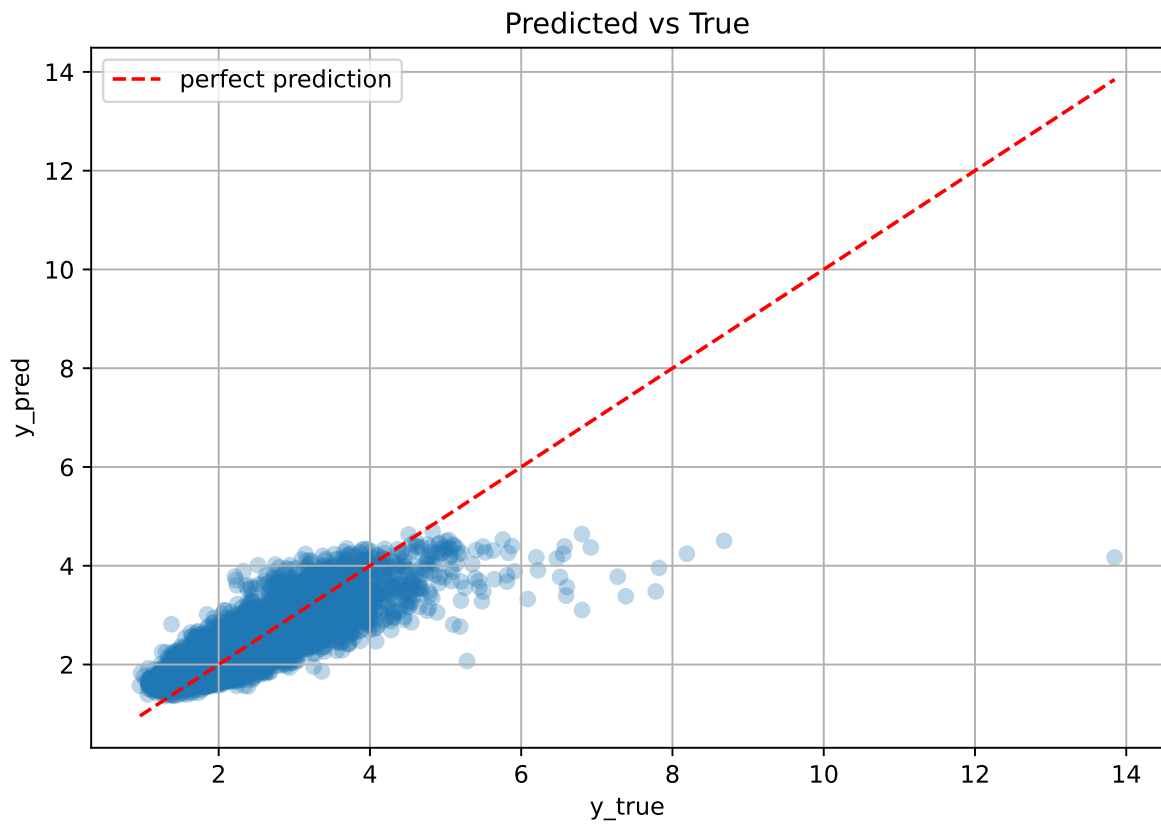
Rank 7: {'reg__n_estimators': 700, 'reg__max_depth': 4, 'reg__learning_rate': 0.014727000915797591, 'reg__num_leaves': 15, 'reg__subsample': 0.6195357798601068, 'reg__colsample_bytree': 0.6425967630461288, 'reg__min_child_samples': 28, 'reg__reg_alpha': 3.7448302438493735, 'reg__reg_lambda': 42.3190254482581, 'reg__min_split_gain': 0.7136807312301099, 'reg__path_smooth': 1.3455776955529741, 'reg__min_child_weight': 0.08958056705201385, 'reg__feature_fraction': 0.7334689444852899, 'reg__bagging_fraction': 0.6655544469628737, 'reg__bagging_freq': 2, 'reg__max_bin': 79, 'reg__min_data_per_group': 199}

Rank 8: {'reg__n_estimators': 700, 'reg__max_depth': 4, 'reg__learning_rate': 0.0144898619253449, 'reg__num_leaves': 16, 'reg__subsample': 0.5480301867369034, 'reg__colsample_bytree': 0.6413021221055409, 'reg__min_child_samples': 27, 'reg__reg_alpha': 3.839768082450142, 'reg__reg_lambda': 18.680162227125894, 'reg__min_split_gain': 0.7982690371211523, 'reg__path_smooth': 1.514849013285871, 'reg__min_child_weight': 0.083464018966645, 'reg__feature_fraction': 0.7345085425225963, 'reg__bagging_fraction': 0.6668372872897903, 'reg__bagging_freq': 2, 'reg__max_bin': 90, 'reg__min_data_per_group': 190}

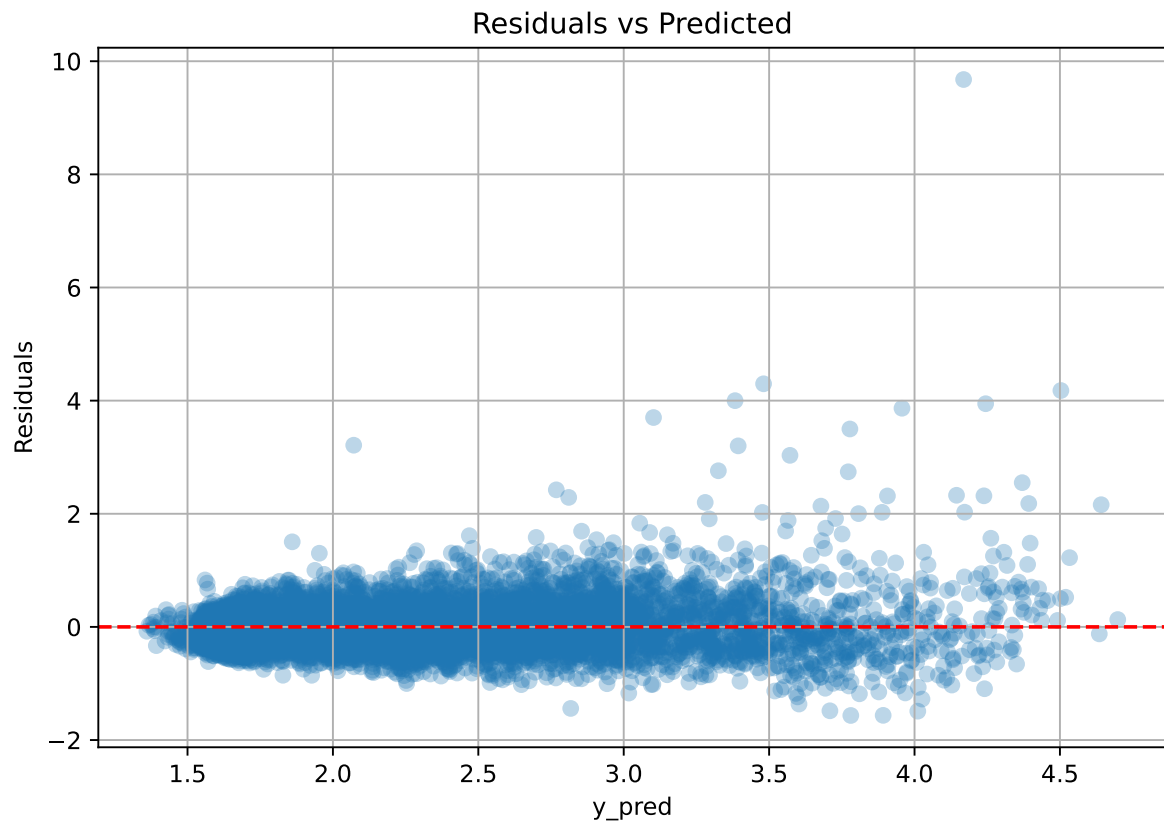
Rank 9: {'reg__n_estimators': 700, 'reg__max_depth': 4, 'reg__learning_rate': 0.015367667072226587, 'reg__num_leaves': 15, 'reg__subsample': 0.6321155093193283, 'reg__colsample_bytree': 0.7163706078890919, 'reg__min_child_samples': 27, 'reg__reg_alpha': 3.6542924451873695, 'reg__reg_lambda': 34.73314081263419, 'reg__min_split_gain': 0.7530236801132792, 'reg__path_smooth': 1.2275700336594866, 'reg__min_child_weight': 0.0890469569169344, 'reg__feature_fraction': 0.5291306614462842, 'reg__bagging_fraction': 0.6721958559465453, 'reg__bagging_freq': 2, 'reg__max_bin': 93, 'reg__min_data_per_group': 198}

Rank 10: {'reg__n_estimators': 800, 'reg__max_depth': 3, 'reg__learning_rate': 0.011675414847677083, 'reg__num_leaves': 16, 'reg__subsample': 0.7183808831608167, 'reg__colsample_bytree': 0.5283716249127278, 'reg__min_child_samples': 49, 'reg__reg_alpha': 0.11050885135011423, 'reg__reg_lambda': 47.43972865926875, 'reg__min_split_gain': 0.47671561171982413, 'reg__path_smooth': 4.694341915106444, 'reg__min_child_weight': 0.06286817927519468, 'reg__feature_fraction': 0.6907818003000281, 'reg__bagging_fraction': 0.5885761069769491, 'reg__bagging_freq': 3, 'reg__max_bin': 108, 'reg__min_data_per_group': 160}

3.5.2 Scatter



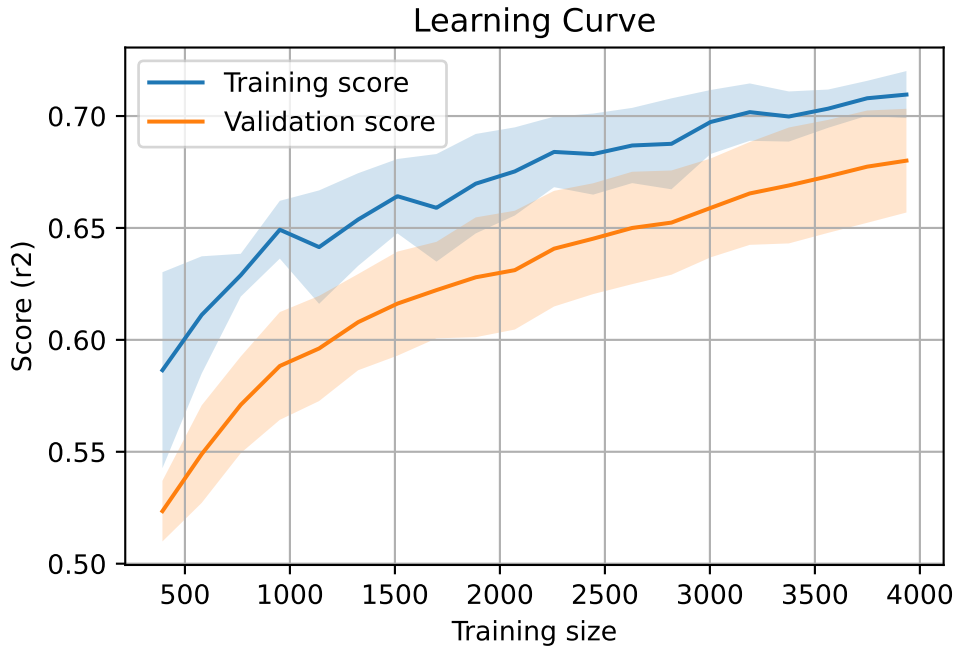
3.5.3 Scatter residuals



3.5.4 Learning curve

`Len(group)` : 8033, `Len(training)` : 5624,

`Len(training_real)` : 3937, `Len(test_of_training)` : 1687, `Len(test)` : 2409

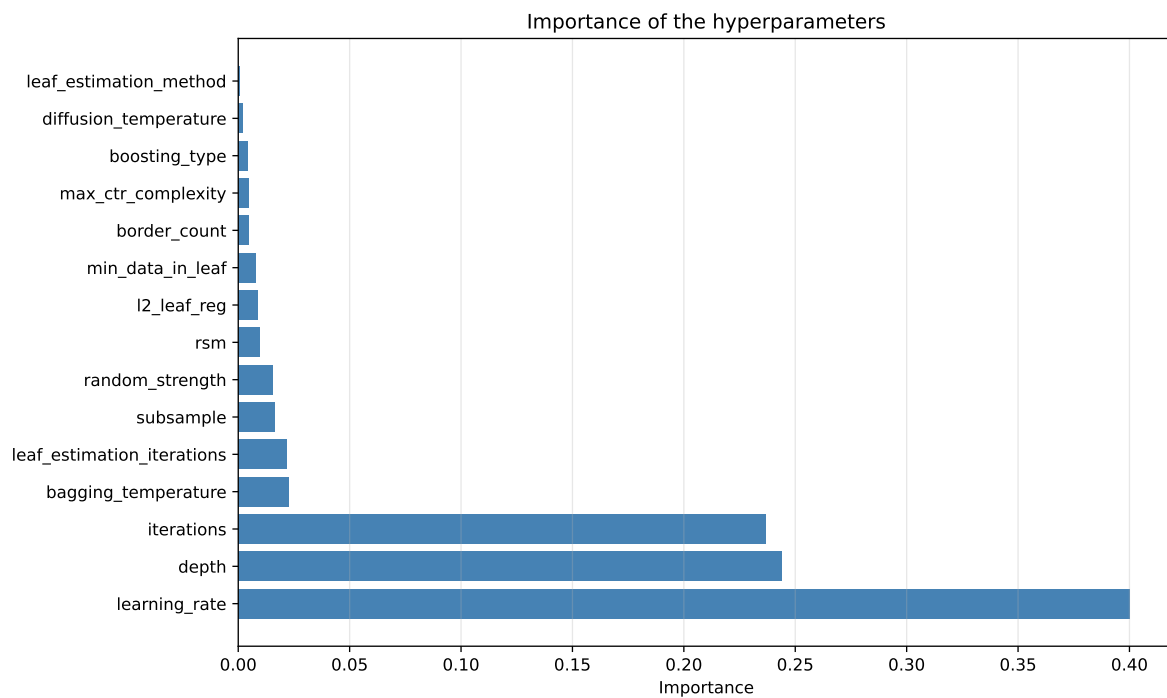
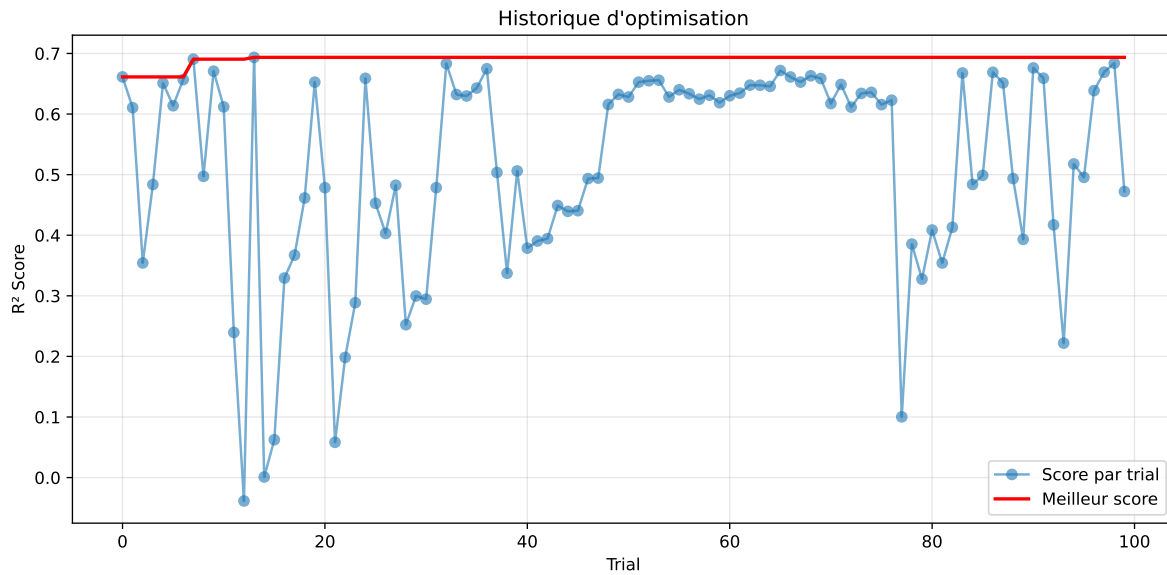


3.6 Model : CatBoost

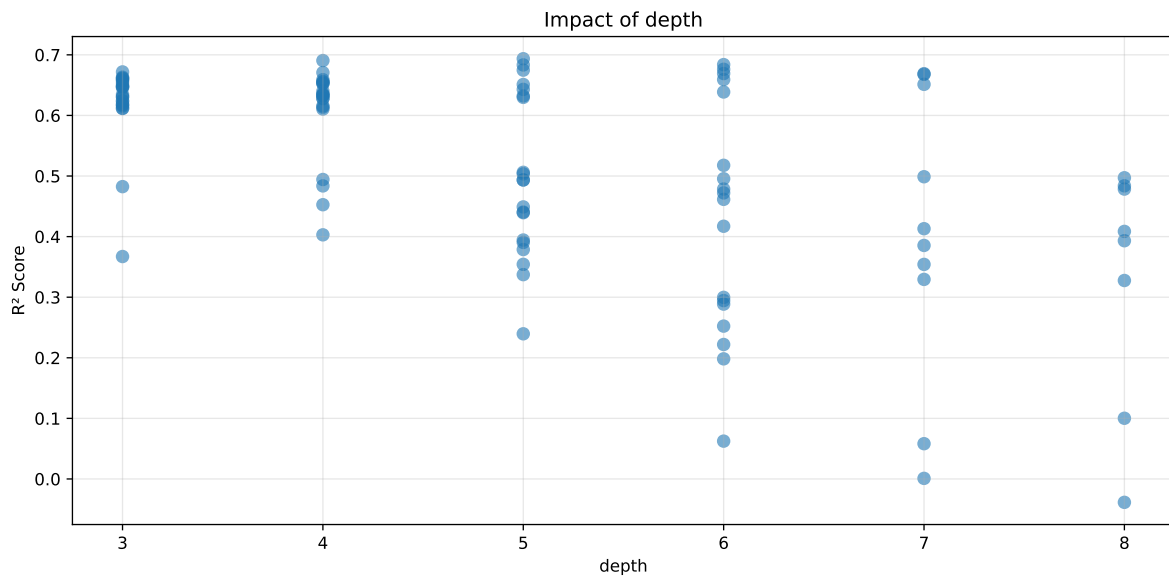
3.6.1 Gridsearch

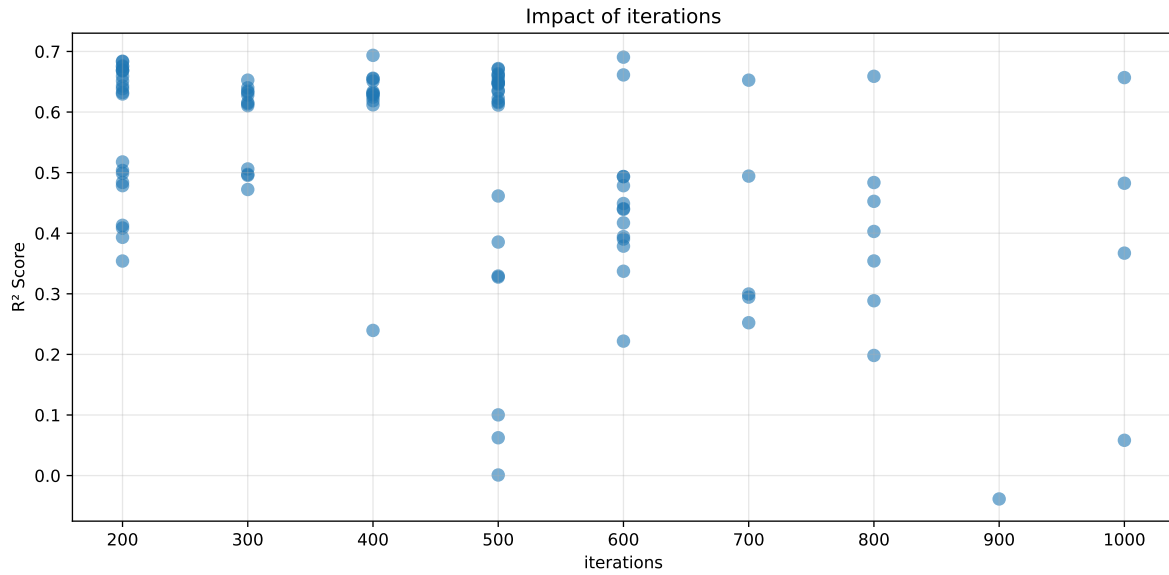
Distribution y_{train} vs y_{test} : y_{train} : mean=2.345, std=0.801, min=0.959, max=13.845
 y_{test} : mean=2.349, std=0.784, min=1.071, max=7.822

===== TOP Importance FEATURES =====
 feature importance r_umidity 20.005779 r_temperature_wl_2h_A_rms 15.566990 r_CO2
 8.867336 r_temperature_wl_2h_A_energy_rel 8.555747 r_temperature_wl_2h_A_energy
 6.636461 r_temperature 6.306483 r_NH3 5.745355 r_NH3_wl_2h_A_energy 5.542037
 r_NH3_wl_2h_D1_energy 4.295108 r_umidity_wl_2h_A_energy 4.201196 r_THI
 3.500564 r_NH3_wl_2h_A_rms 3.389272 r_umidity_wl_2h_D1_energy_rel 2.672234
 r_NH3_wl_2h_A_energy_rel 2.511744 r_umidity_wl_2h_A_rms 2.203692



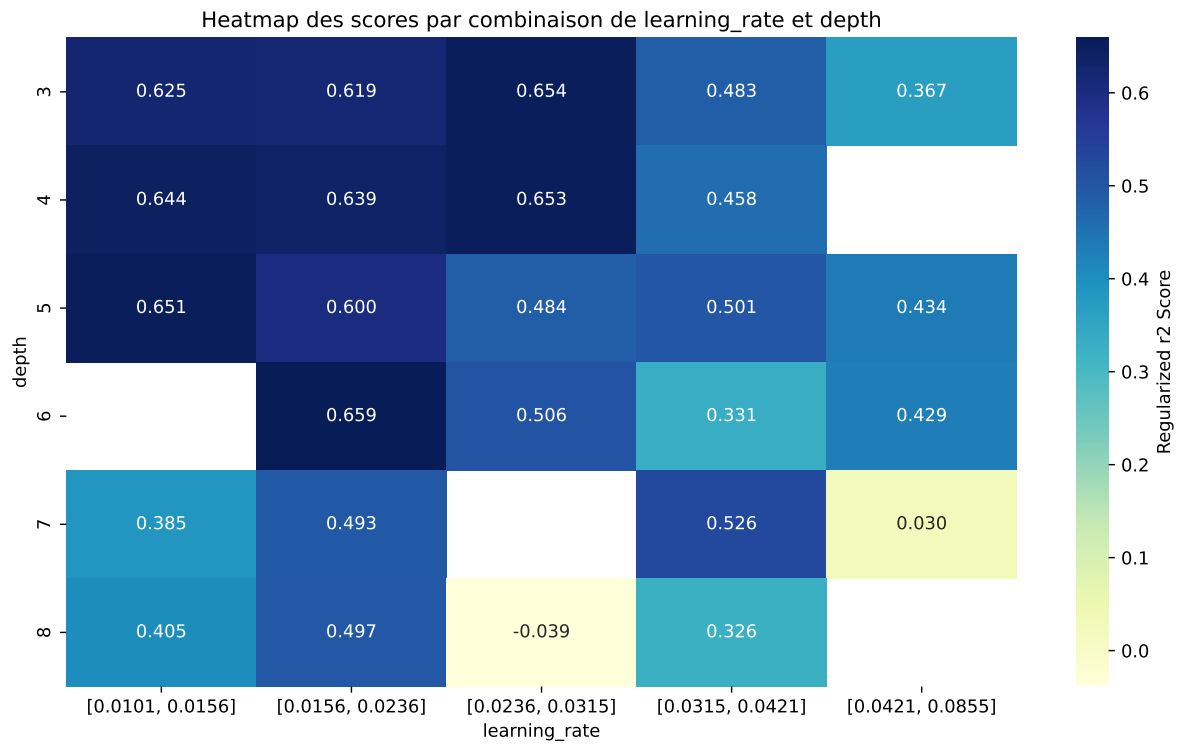
Paramètres avec >5% d'importance : ['learning_rate', 'depth', 'iterations']





C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.7261	0.6936	0.6631	0.0247	0.6936	0.6701	-0.0234	0.9662	1.0469
2	0.7201	0.6906	0.6623	0.0224	0.6906	0.6674	-0.0231	0.9665	1.0428
3	0.7138	0.6839	0.654	0.023	0.6839	0.6614	-0.0225	0.9671	1.0437
4	0.7101	0.6833	0.6556	0.0227	0.6833	0.6631	-0.0202	0.9704	1.0392
5	0.7028	0.6761	0.6444	0.0221	0.6761	0.652	-0.0241	0.9643	1.0395
6	0.6997	0.6748	0.6467	0.0232	0.6748	0.654	-0.0208	0.9692	1.037
7	0.6925	0.6718	0.6428	0.0232	0.6718	0.6467	-0.0252	0.9626	1.0308
8	0.6971	0.6707	0.6403	0.0233	0.6707	0.6447	-0.026	0.9612	1.0393
9	0.6929	0.6691	0.6397	0.0221	0.6691	0.6434	-0.0258	0.9615	1.0355
10	0.6956	0.6689	0.6378	0.0243	0.6689	0.6514	-0.0175	0.9738	1.0398
---	---	---	---	---	---	---	---	---	---
100	0.9494	0.7847	0.7536	0.0297	-0.0386	0.7658	-0.0189	0.9759	1.2098

Hyperparameters:

Rank 1: {'reg__iterations': 400, 'reg__depth': 5, 'reg__learning_rate': 0.026042923212587134, 'reg__l2_leaf_reg': 47.492659093301974, 'reg__bagging_temperature': 2.209176859075569, 'reg__random_strength': 2.135417113003788, 'reg__subsample': 0.6105288003339664, 'reg__min_data_in_leaf': 38, 'reg__leaf_estimation_iterations': 6, 'reg__rsm': 0.5449499652888313, 'reg__border_count': 35, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 8531.757947601023}

Rank 2: {'reg__iterations': 600, 'reg__depth': 4, 'reg__learning_rate': 0.019718084286149243, 'reg__l2_leaf_reg': 6.190037879750024, 'reg__bagging_temperature': 3.0703074254954488, 'reg__random_strength': 1.352904157880357, 'reg__subsample': 0.883625166861326, 'reg__min_data_in_leaf': 47, 'reg__leaf_estimation_iterations': 9, 'reg__rsm': 0.5760599467623255, 'reg__border_count': 114, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 2345.2663079610047}

Rank 3: {'reg__iterations': 200, 'reg__depth': 6, 'reg__learning_rate': 0.047115155555593057, 'reg__l2_leaf_reg': 42.343585487545674, 'reg__bagging_temperature': 2.5234304283275133, 'reg__random_strength': 3.419066974074812, 'reg__subsample': 0.6179047255419191, 'reg__min_data_in_leaf': 13, 'reg__leaf_estimation_iterations': 5, 'reg__rsm': 0.5449322983193547, 'reg__border_count': 113, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 5014.684984886634}

Rank 4: {'reg__iterations': 200, 'reg__depth': 5, 'reg__learning_rate': 0.04389895738082638, 'reg__l2_leaf_reg': 38.36193628401847, 'reg__bagging_temperature': 2.53833258473502, 'reg__random_strength': 0.862400562881702, 'reg__subsample': 0.8975925989801093, 'reg__min_data_in_leaf': 44, 'reg__leaf_estimation_iterations': 5, 'reg__rsm':

0.5053698501305788, 'reg__border_count': 32, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 6124.38204847845}

Rank 5: {'reg__iterations': 200, 'reg__depth': 6, 'reg__learning_rate': 0.04478649744233711, 'reg__l2_leaf_reg': 42.953936471717036, 'reg__bagging_temperature': 2.6450386212752326, 'reg__random_strength': 3.4335714428886743, 'reg__subsample': 0.6193490695083782, 'reg__min_data_in_leaf': 13, 'reg__leaf_estimation_iterations': 2, 'reg__rsm': 0.5446984973641364, 'reg__border_count': 113, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 4723.660701170358}

Rank 6: {'reg__iterations': 200, 'reg__depth': 5, 'reg__learning_rate': 0.032982879016905074, 'reg__l2_leaf_reg': 20.53781469010918, 'reg__bagging_temperature': 3.1624918694646134, 'reg__random_strength': 0.6753293375161327, 'reg__subsample': 0.8958414800045421, 'reg__min_data_in_leaf': 50, 'reg__leaf_estimation_iterations': 5, 'reg__rsm': 0.5453168024113328, 'reg__border_count': 68, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 5840.448851152061}

Rank 7: {'reg__iterations': 500, 'reg__depth': 3, 'reg__learning_rate': 0.026686826699980615, 'reg__l2_leaf_reg': 42.44284265350059, 'reg__bagging_temperature': 3.9766069142670237, 'reg__random_strength': 0.805842162688794, 'reg__subsample': 0.758788865550881, 'reg__min_data_in_leaf': 18, 'reg__leaf_estimation_iterations': 3, 'reg__rsm': 0.5897016424173362, 'reg__border_count': 92, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 1455.5011440841454}

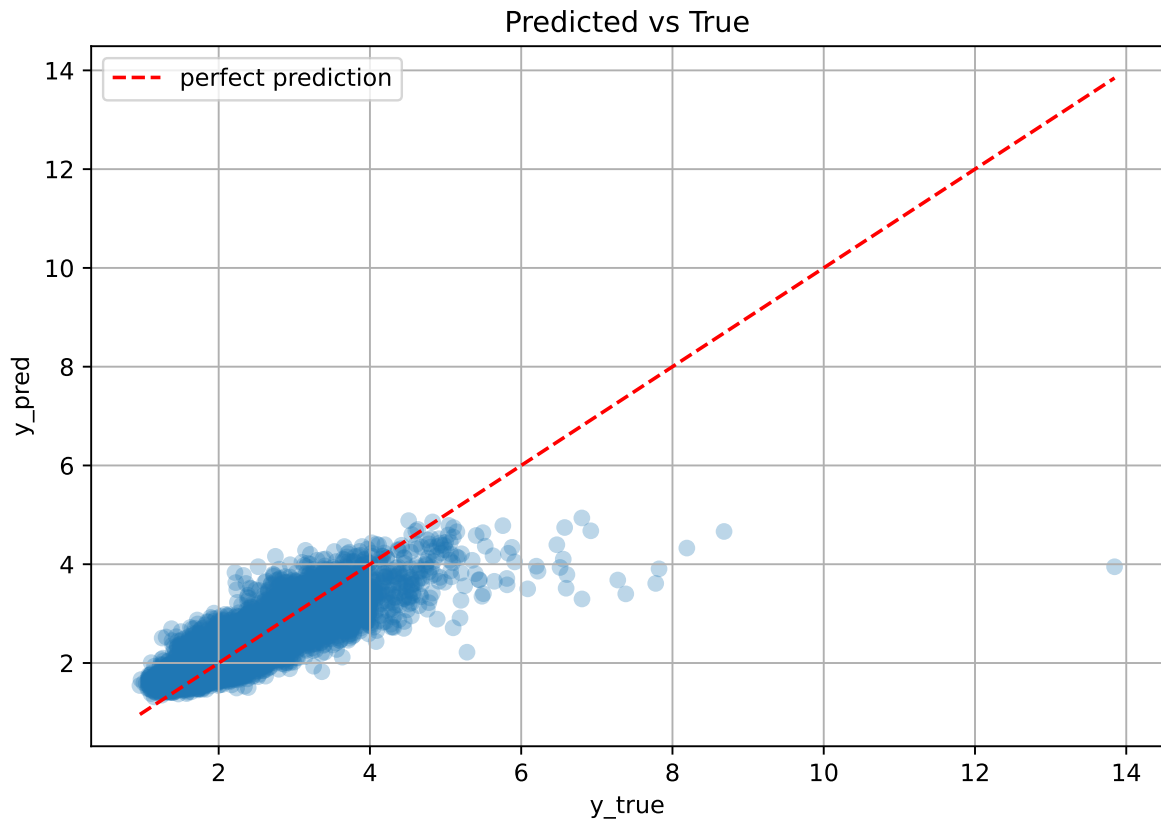
Rank 8: {'reg__iterations': 500, 'reg__depth': 4, 'reg__learning_rate': 0.013997286281408308, 'reg__l2_leaf_reg': 45.54214513943148, 'reg__bagging_temperature': 4.855117945777479, 'reg__random_strength': 0.10707556795730544, 'reg__subsample': 0.5149656726519807, 'reg__min_data_in_leaf': 18, 'reg__leaf_estimation_iterations': 7, 'reg__rsm': 0.7385110290052297, 'reg__border_count': 51, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 8202.762370198865}

Rank 9: {'reg__iterations': 200, 'reg__depth': 6, 'reg__learning_rate': 0.04399464574738027, 'reg__l2_leaf_reg': 42.2226914789688, 'reg__bagging_temperature': 2.5500706736346768, 'reg__random_strength': 3.498349871235253, 'reg__subsample': 0.6335825467683952, 'reg__min_data_in_leaf': 23, 'reg__leaf_estimation_iterations': 3, 'reg__rsm': 0.5449060587288316, 'reg__border_count': 123, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 511.4078865123199}

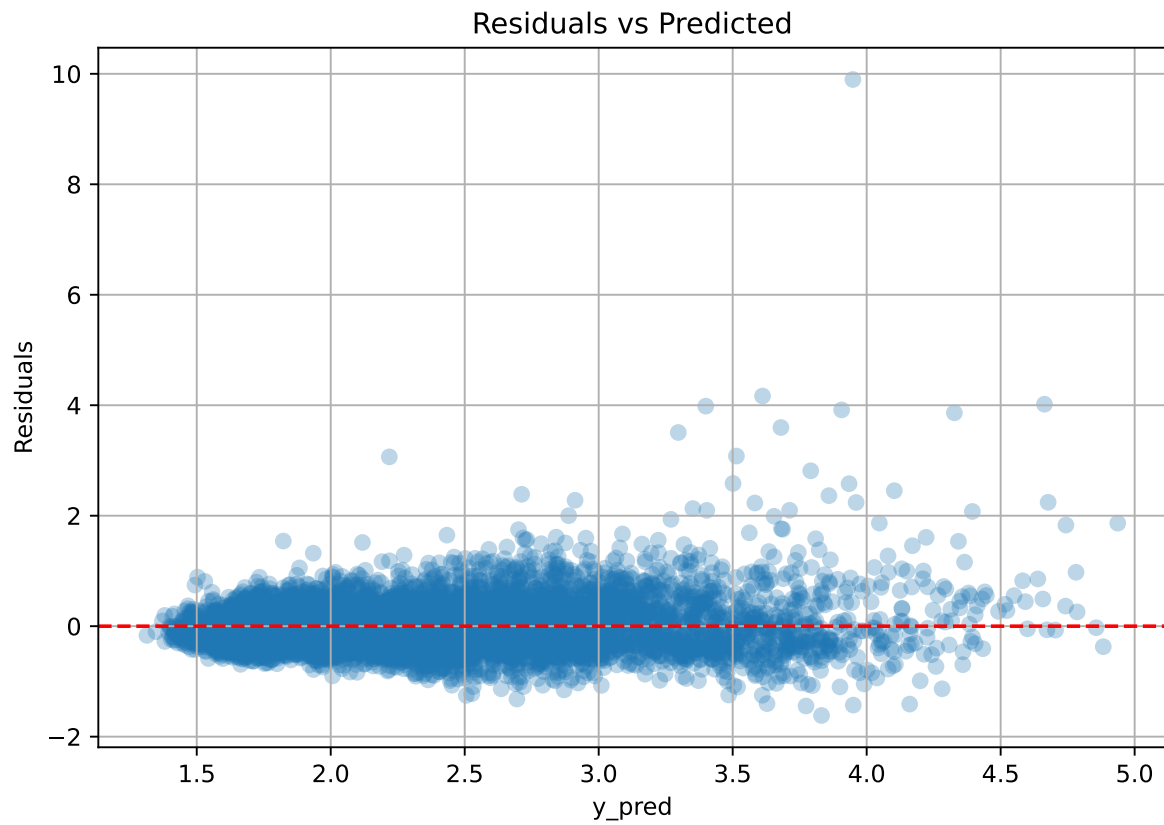
Rank 10: {'reg__iterations': 200, 'reg__depth': 7, 'reg__learning_rate': 0.035405726850022974, 'reg__l2_leaf_reg': 43.16164522451944, 'reg__bagging_temperature': 2.6798767335512705,

'reg__random_strength': 3.4889331143362594, 'reg__subsample': 0.6927668484511076,
'reg__min_data_in_leaf': 25, 'reg__leaf_estimation_iterations': 2, 'reg__rsm':
0.5191859052340285, 'reg__border_count': 117, 'reg__leaf_estimation_method': 'Gradient',
'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature':
4725.773873672667}

3.6.2 Scatter



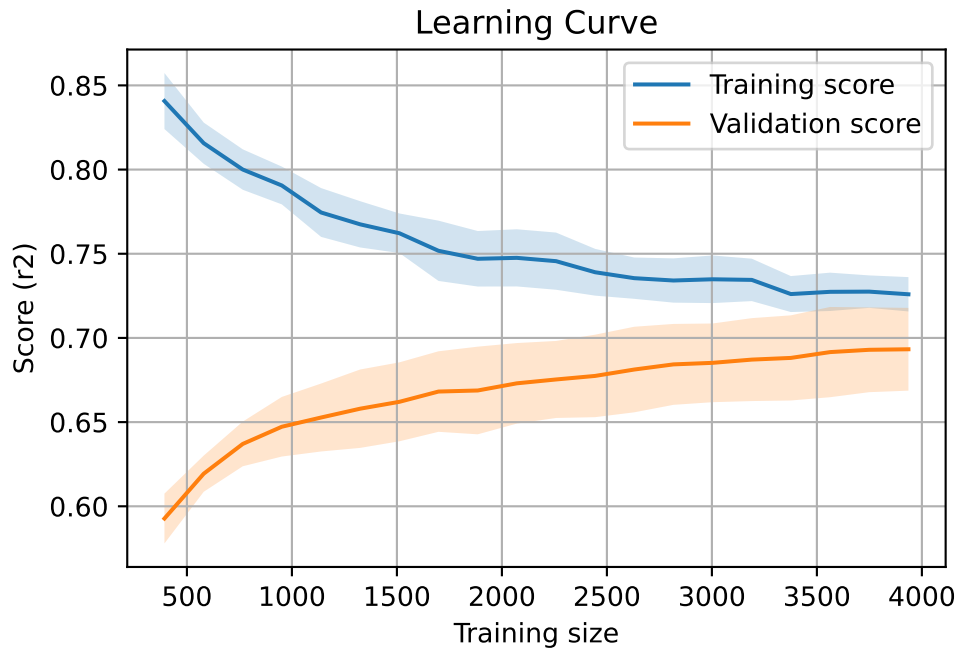
3.6.3 Scatter residuals



3.6.4 Learning curve

`Len(group)` : 8033, `Len(training)` : 5624,

`Len(training_real)` : 3937, `Len(test_of_training)` : 1687, `Len(test)` : 2409



3.7 Sumup-table

model	mean_train_cv	mean_test_cv	std_test_cv	test_cv	test	diff	r_test	r_train	pen_score
Linear	0.517	0.535	0.02	0.495	0.495	-0.039	0.927	0.967	nan
LGBM	0.712	0.682	0.024	0.651	0.672	-0.01	0.986	1.044	0.6818
CatBoost	0.726	0.694	0.025	0.663	0.67	-0.023	0.966	1.047	0.6936

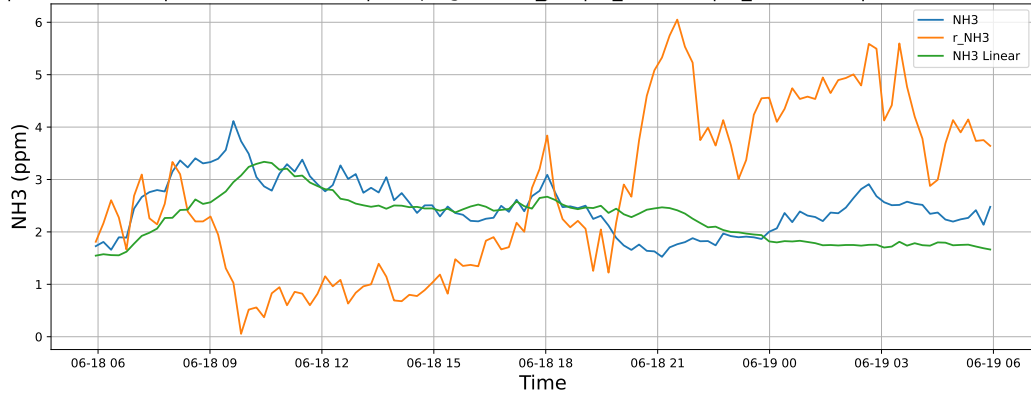
4 Plot

4.1 Standalone plots

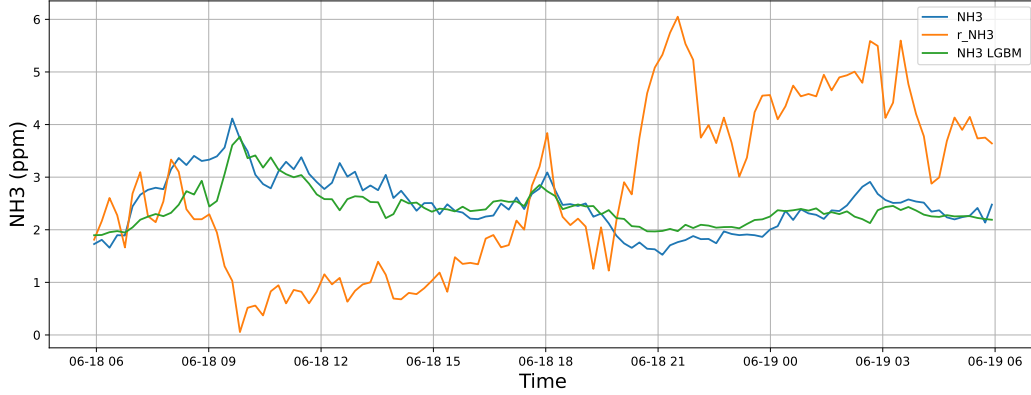
4.1.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

4.1.1.1 Time window : 24

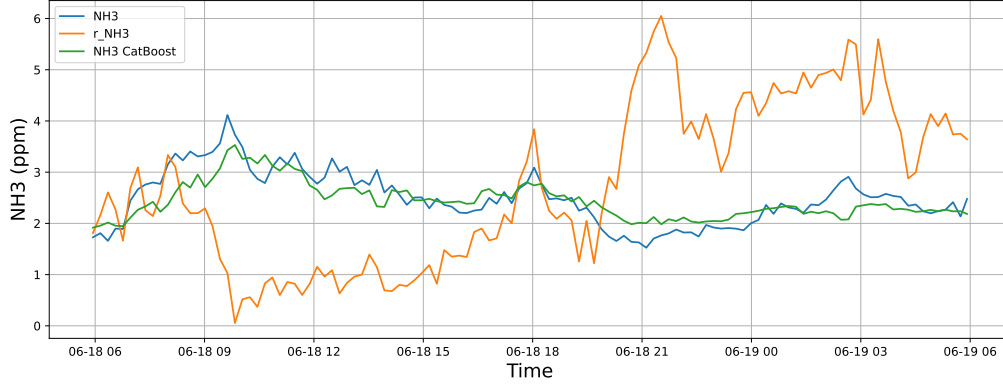
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

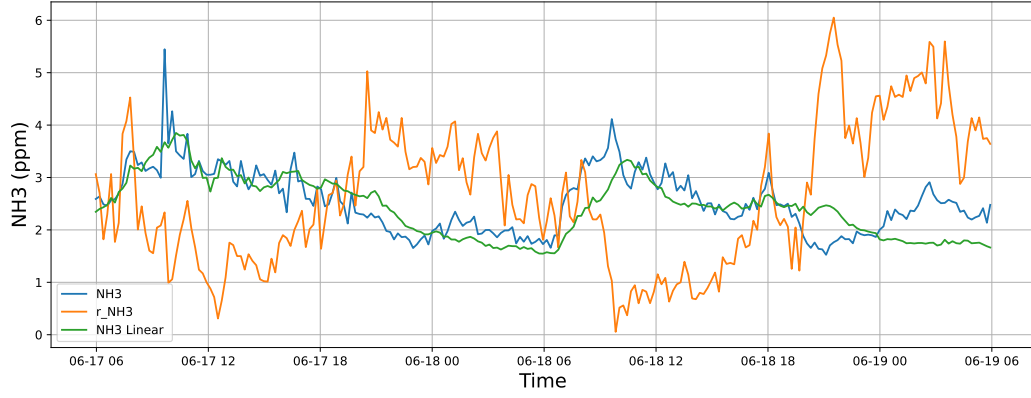


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

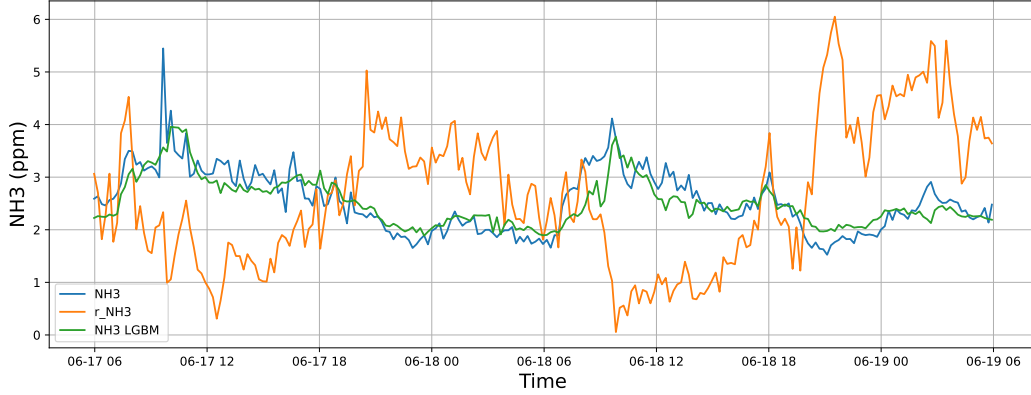


4.1.1.2 Time window : 48

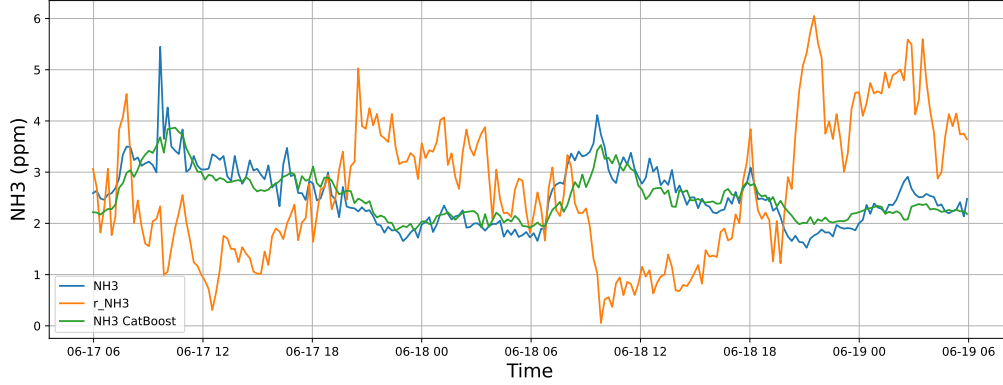
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

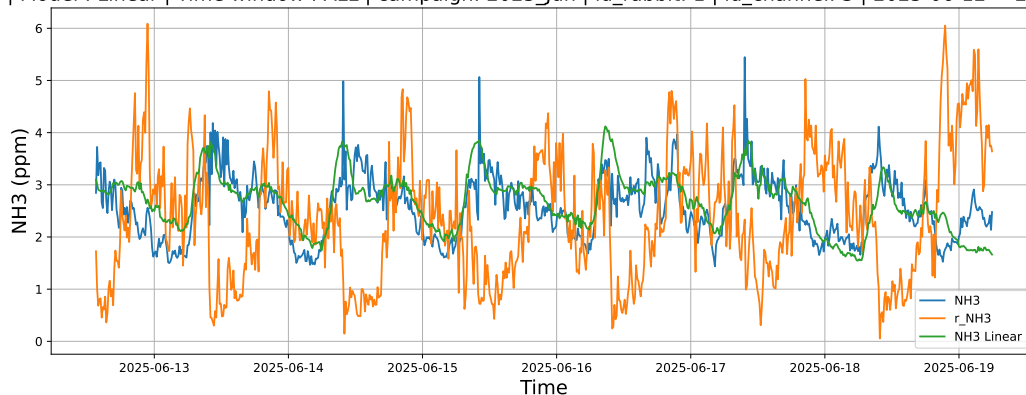


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

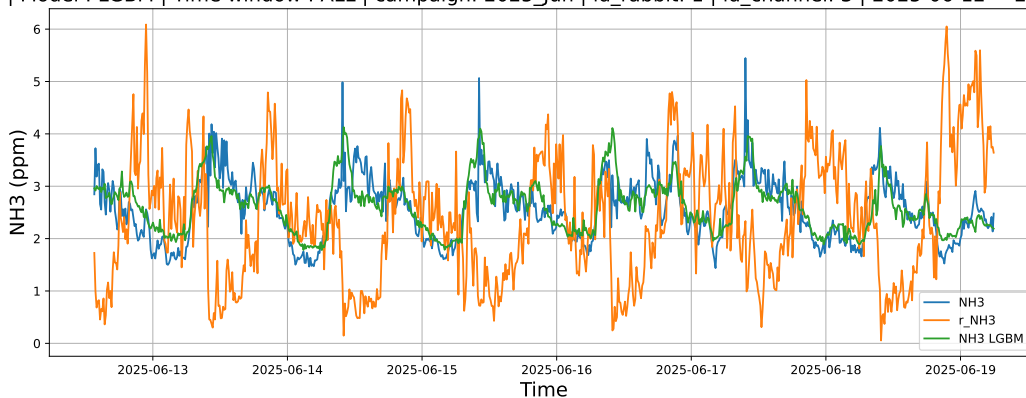


4.1.1.3 Time window : ALL

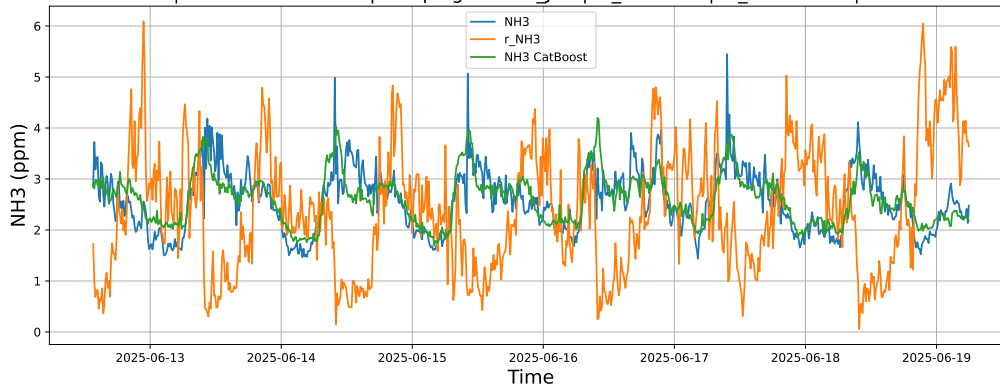
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



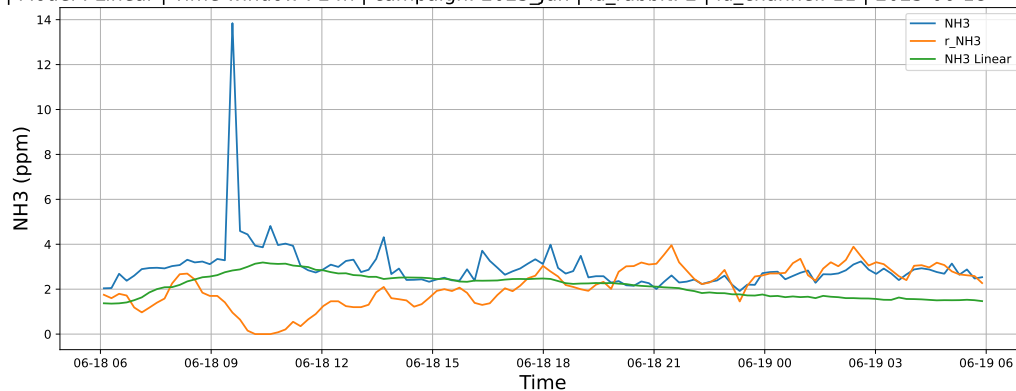
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



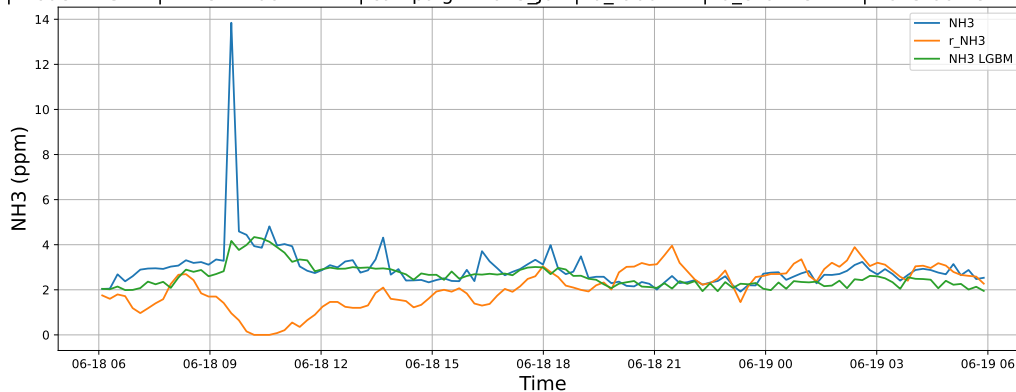
4.1.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

4.1.2.1 Time window : 24

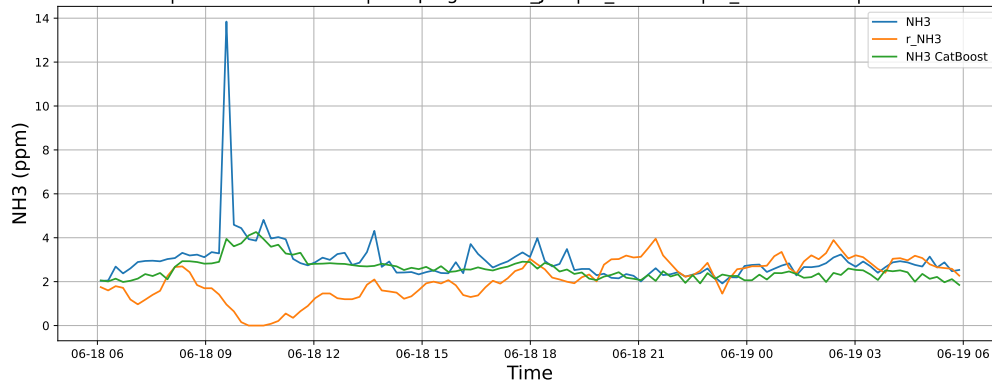
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

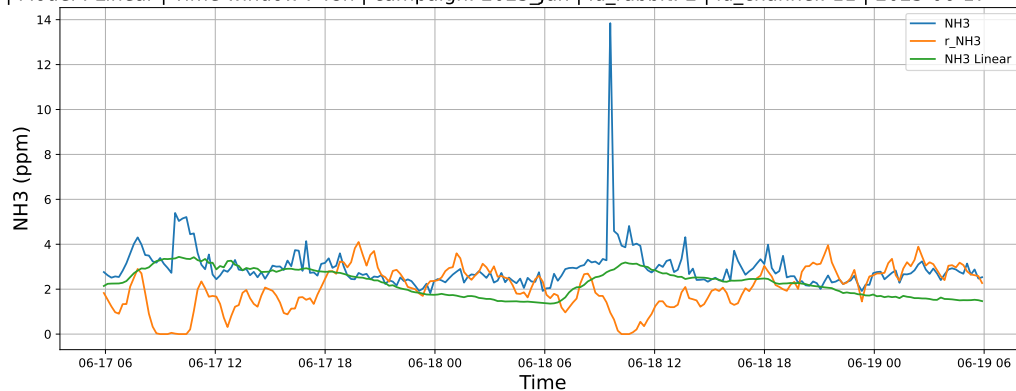


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

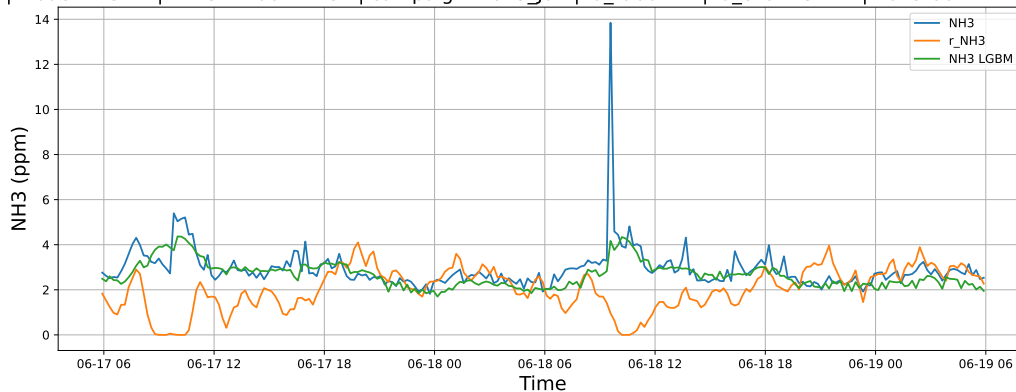


4.1.2.2 Time window : 48

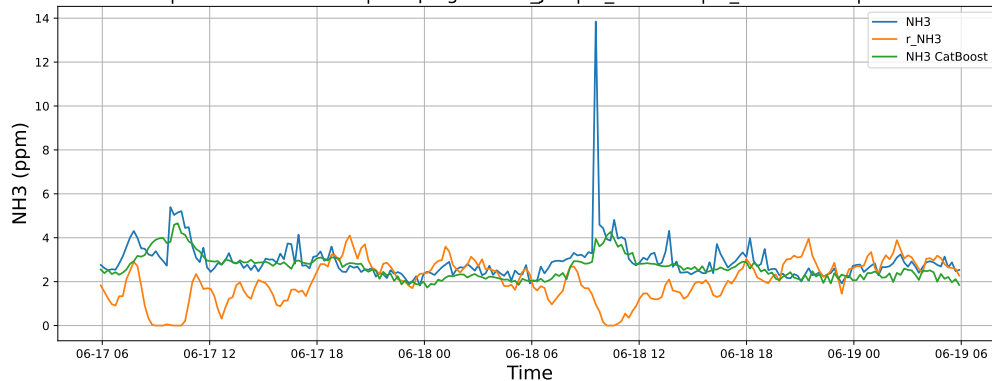
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

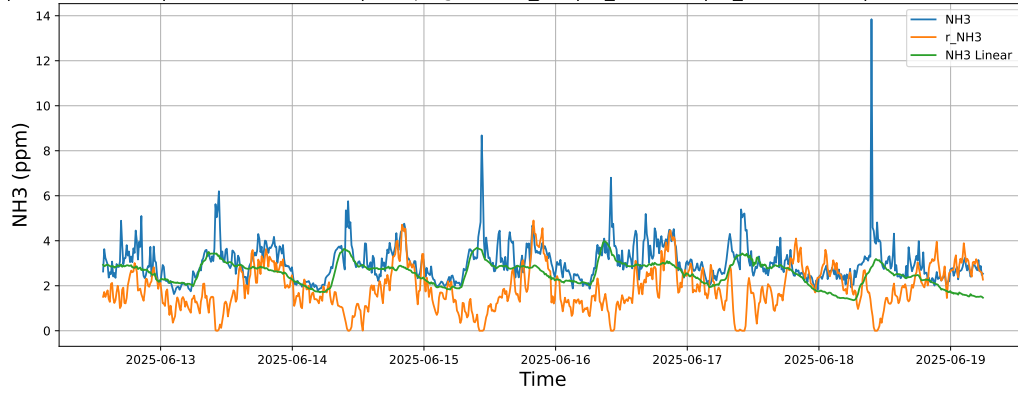


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

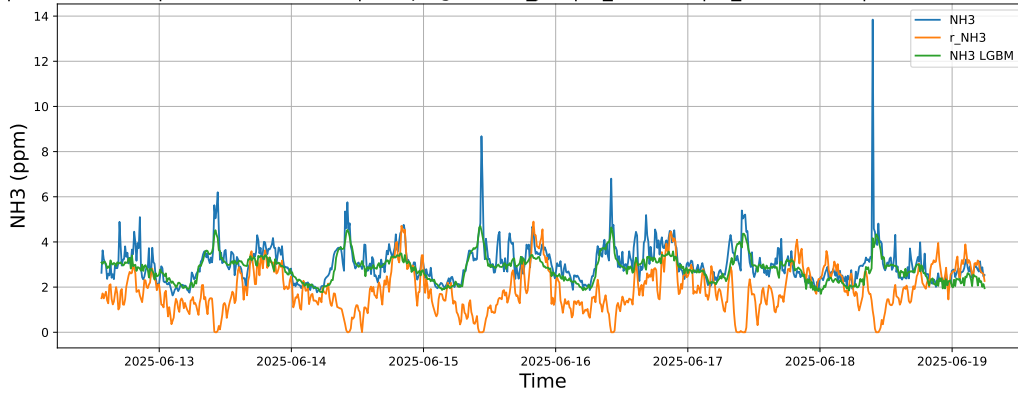


4.1.2.3 Time window : ALL

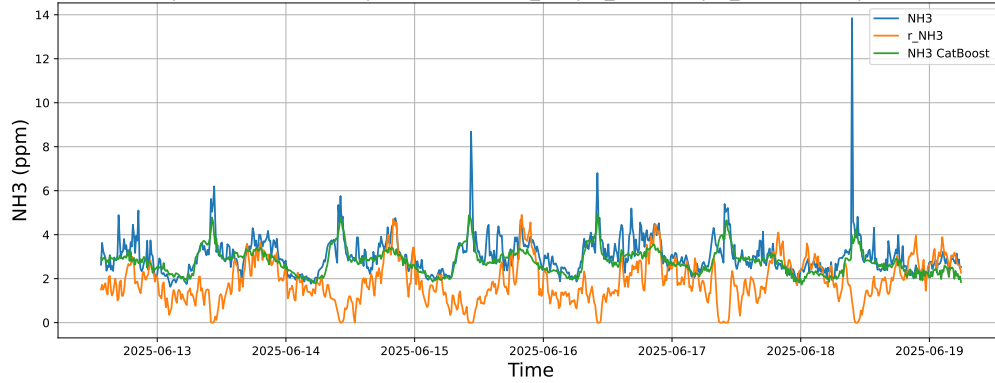
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



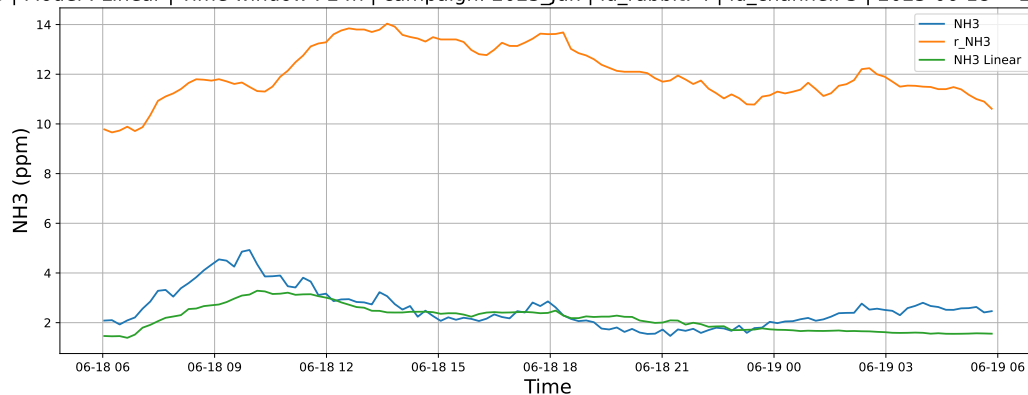
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



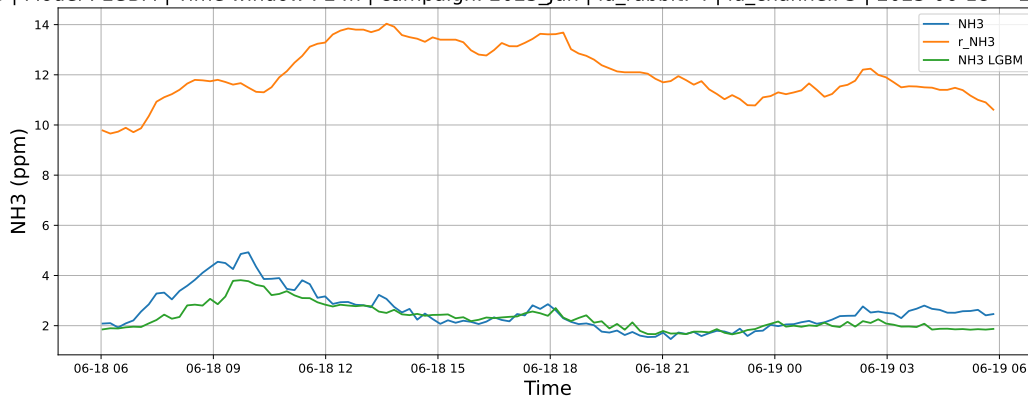
4.1.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

4.1.3.1 Time window : 24

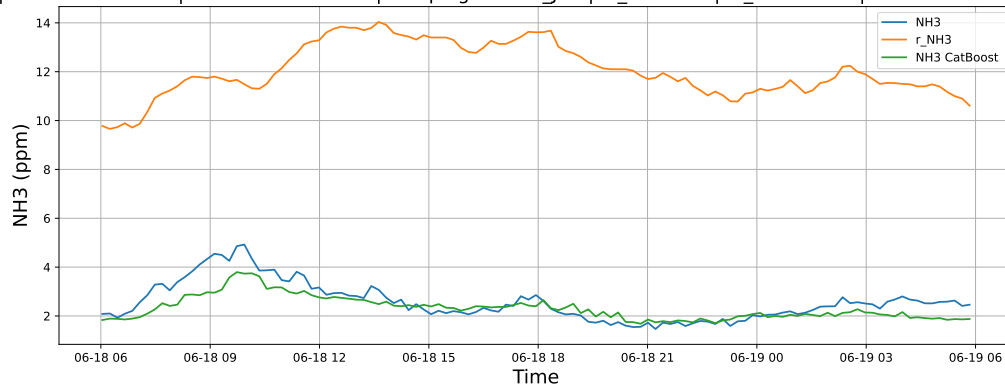
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

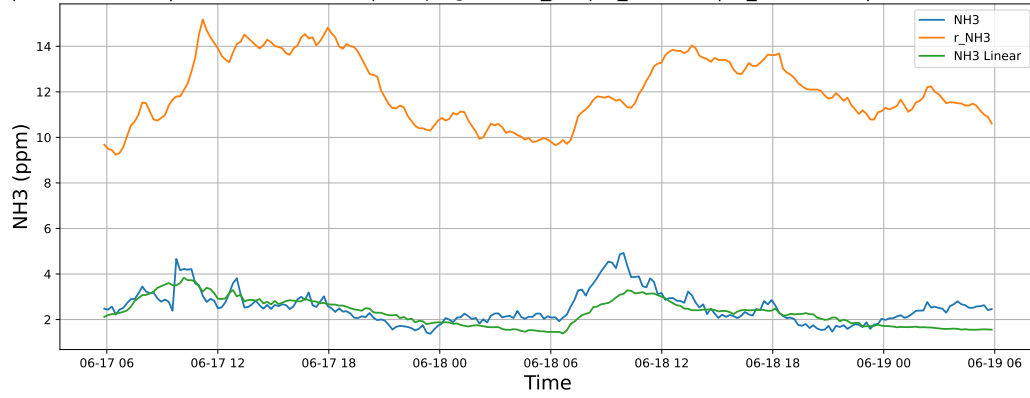


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

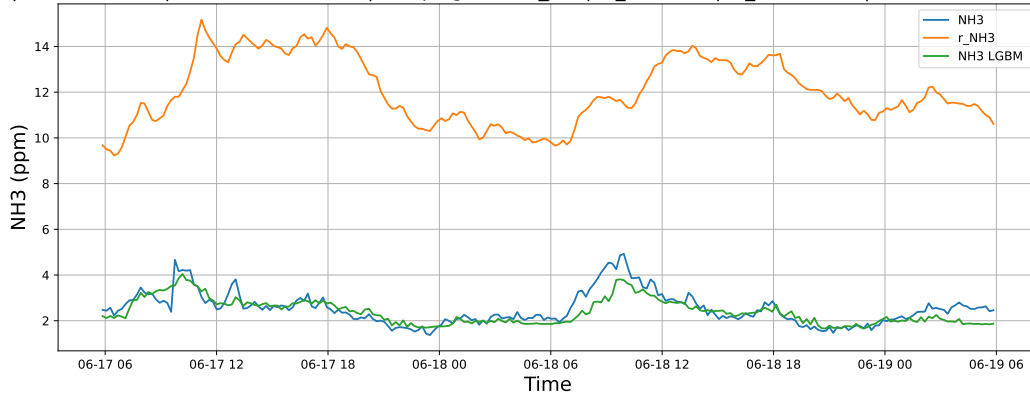


4.1.3.2 Time window : 48

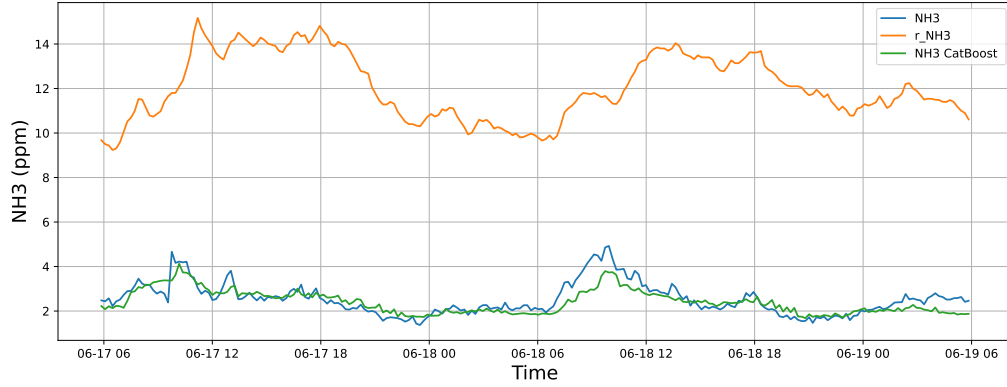
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

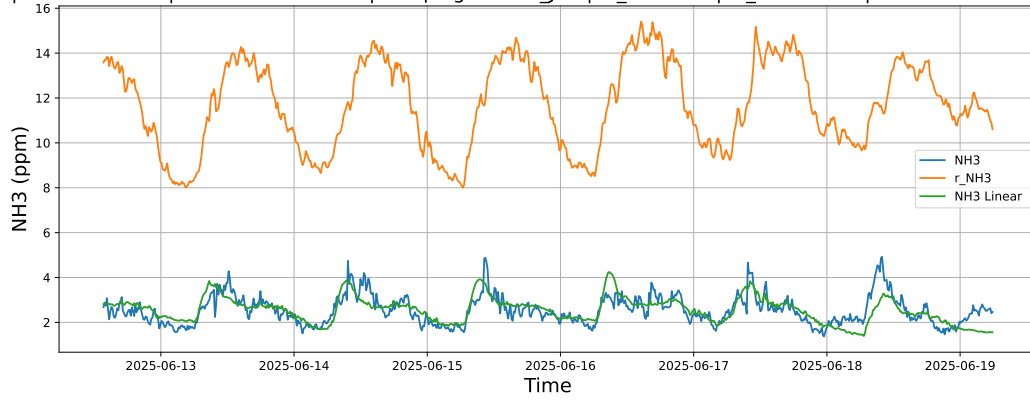


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

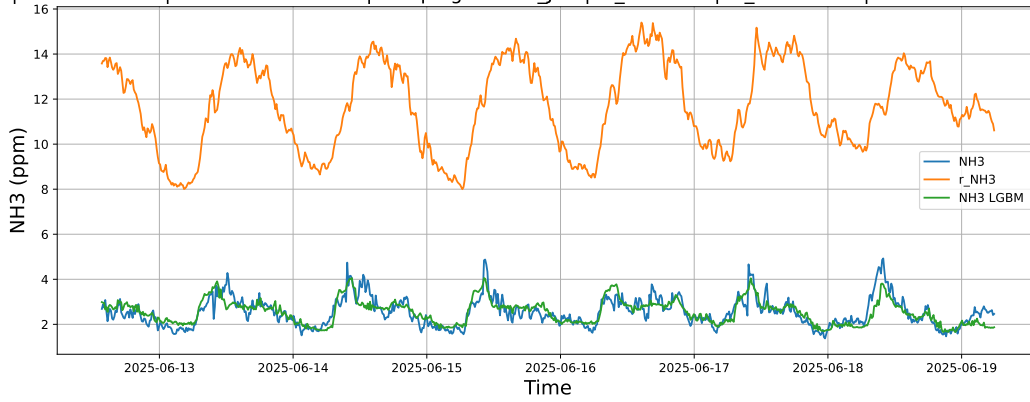


4.1.3.3 Time window : ALL

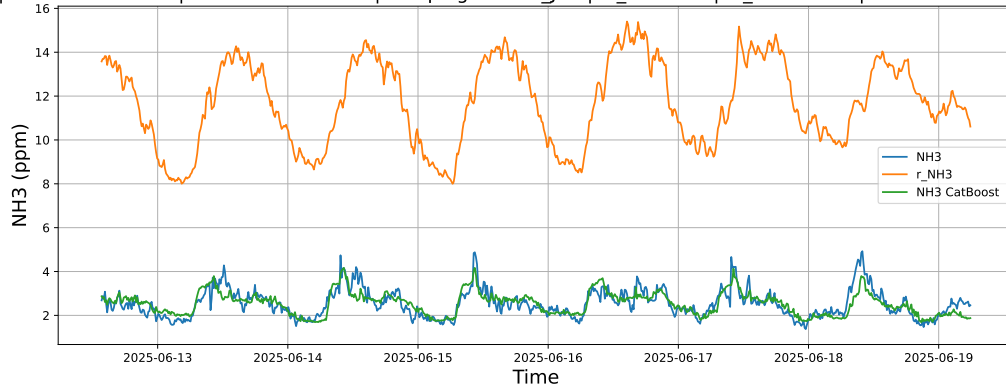
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



4.1.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

4.1.4.1 Time window : 24

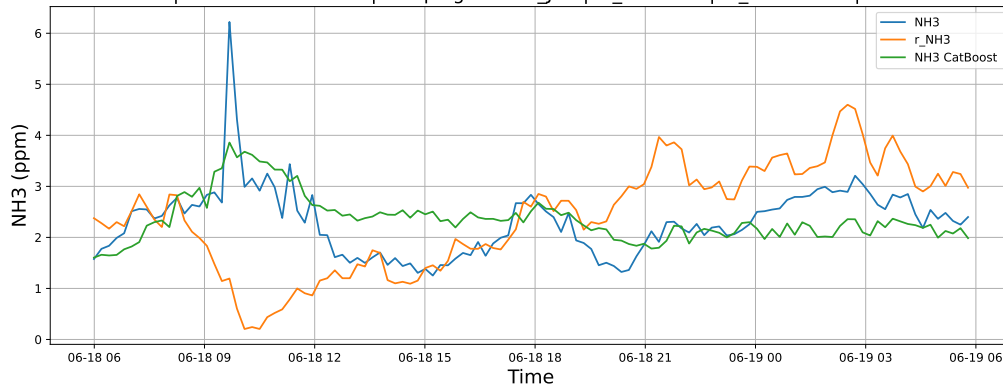
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

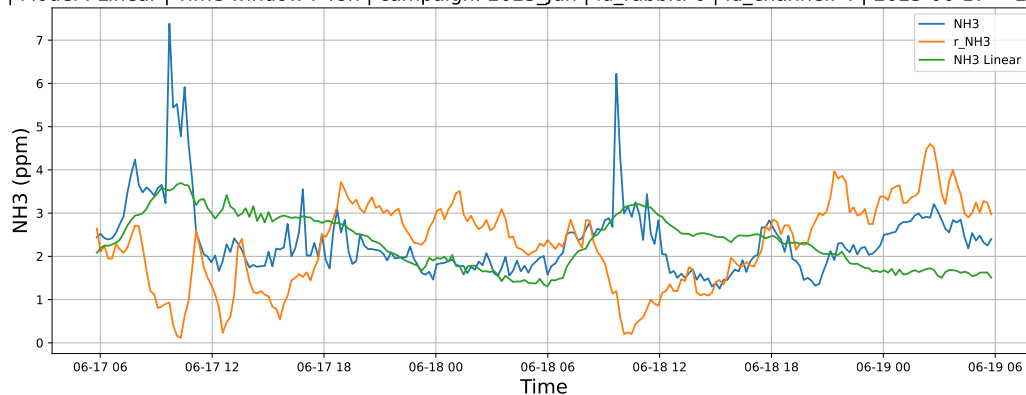


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

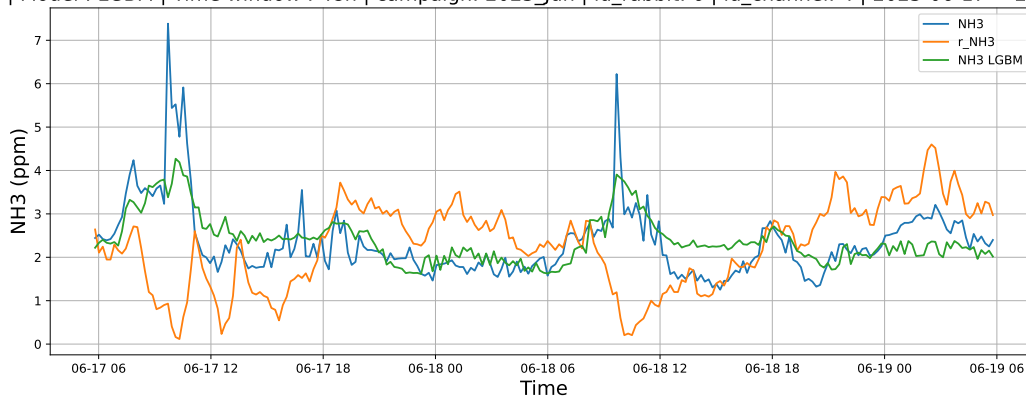


4.1.4.2 Time window : 48

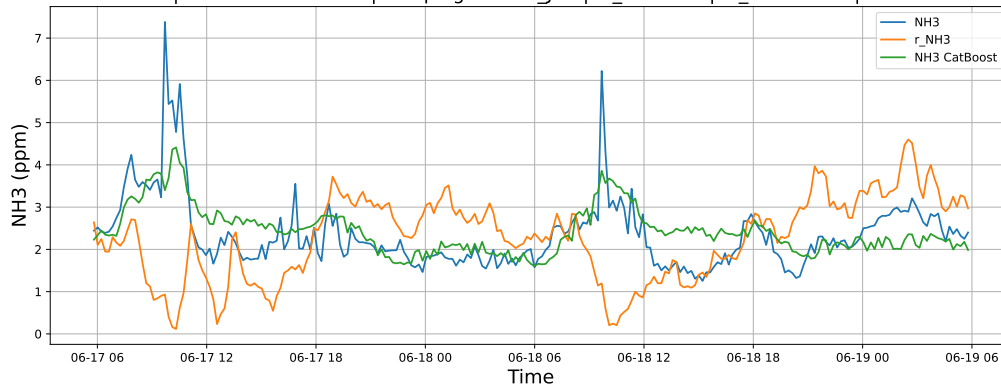
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

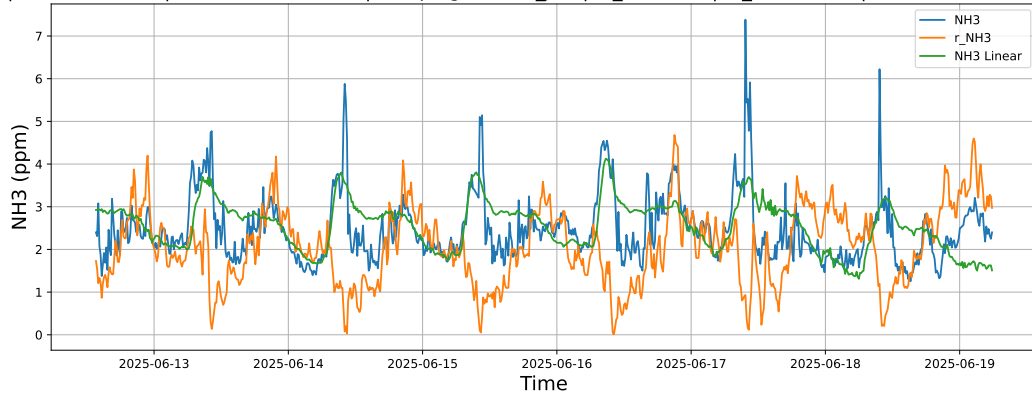


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

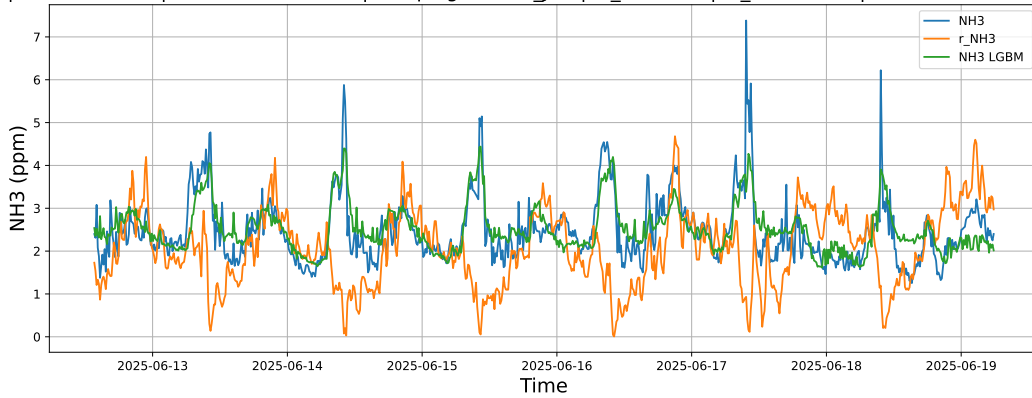


4.1.4.3 Time window : ALL

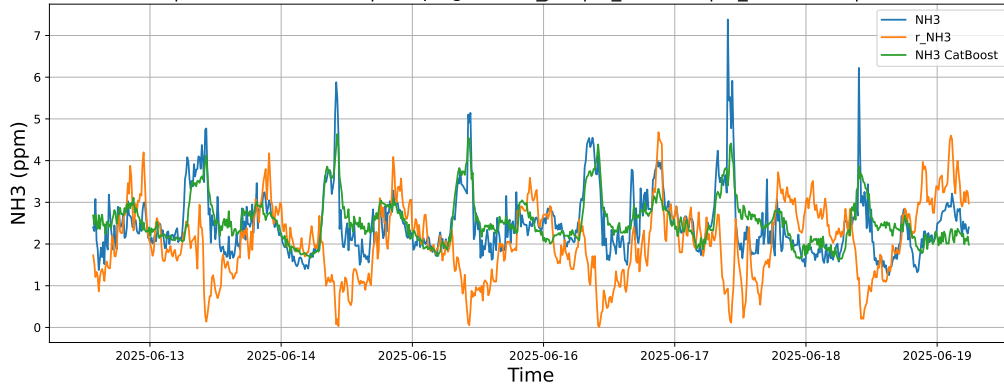
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



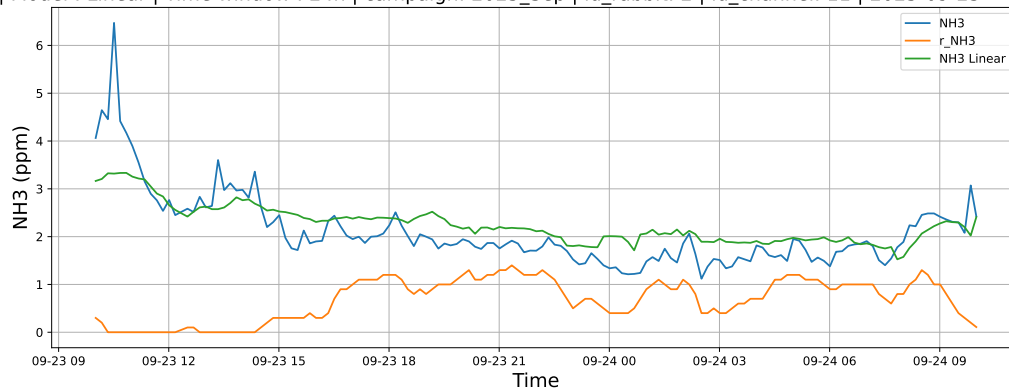
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



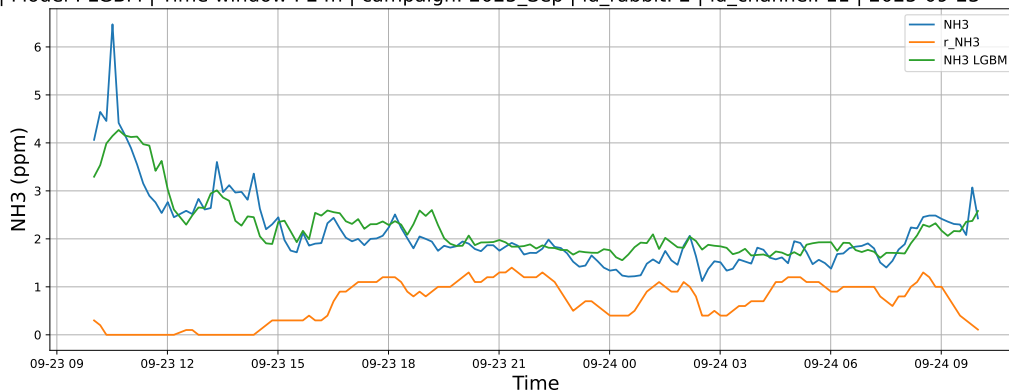
4.1.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

4.1.5.1 Time window : 24

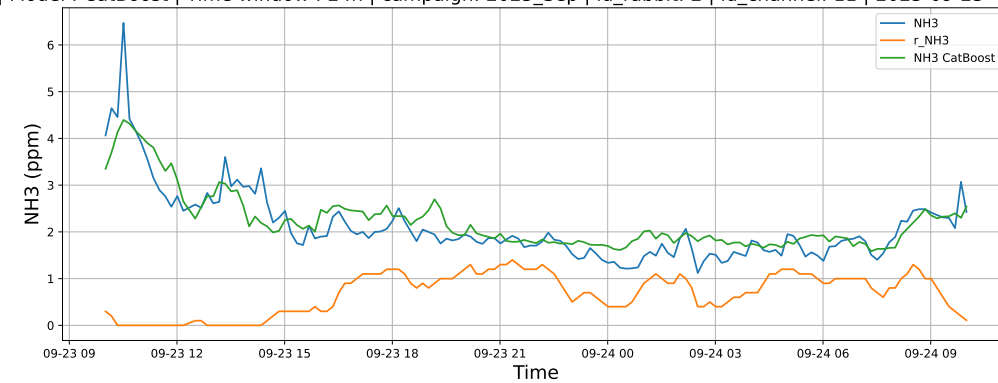
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

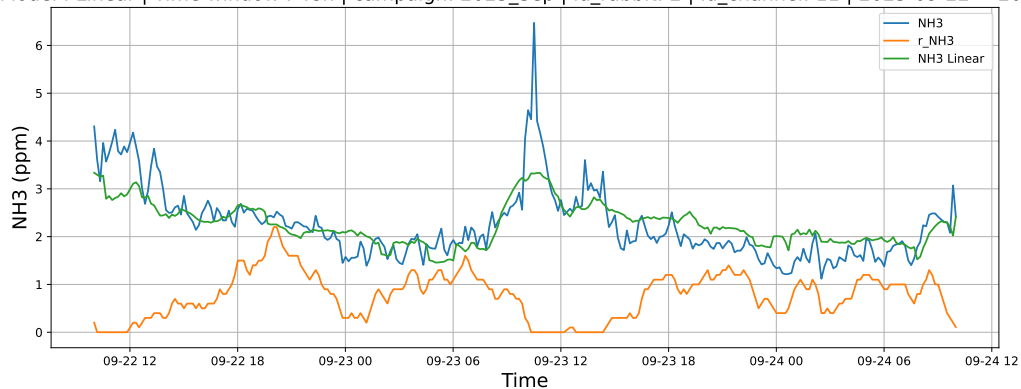


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

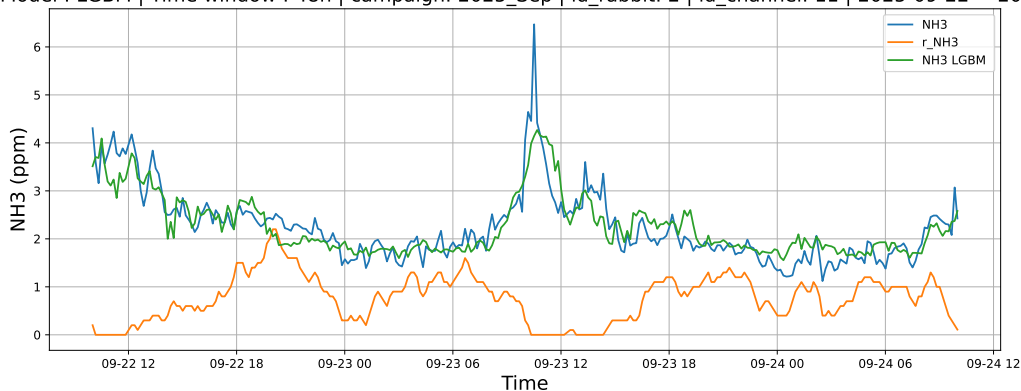


4.1.5.2 Time window : 48

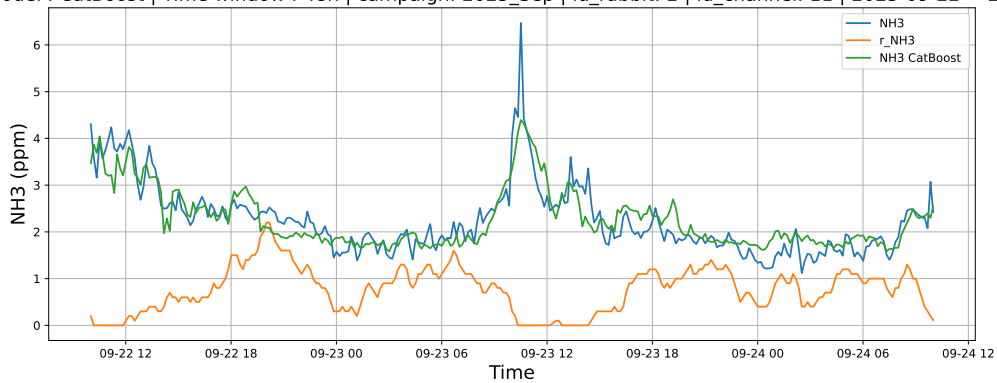
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

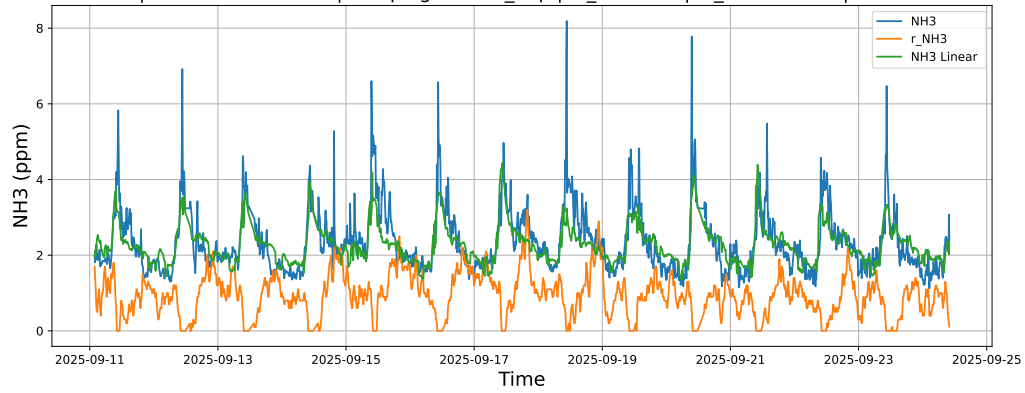


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

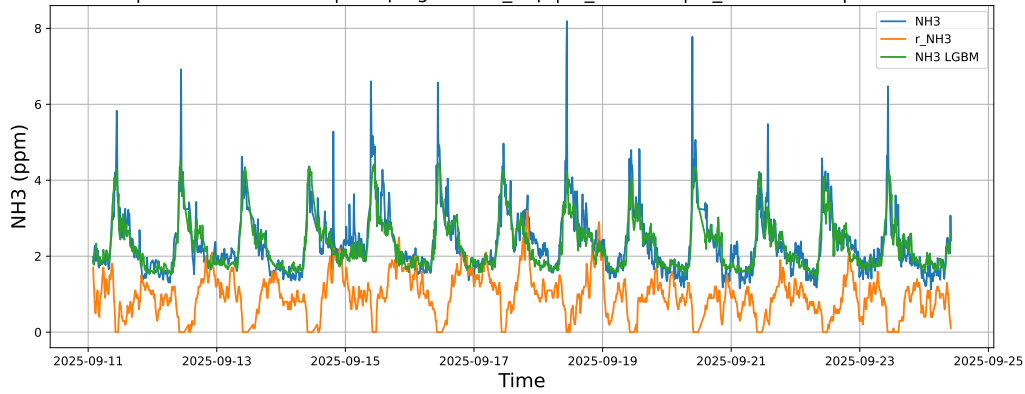


4.1.5.3 Time window : ALL

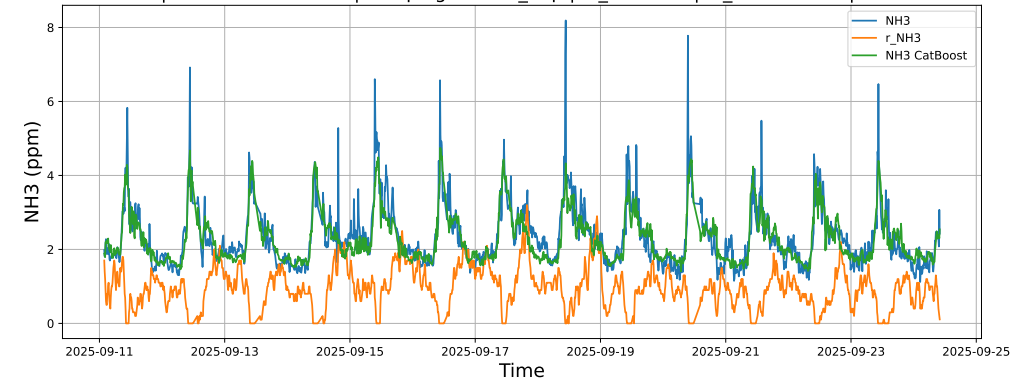
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



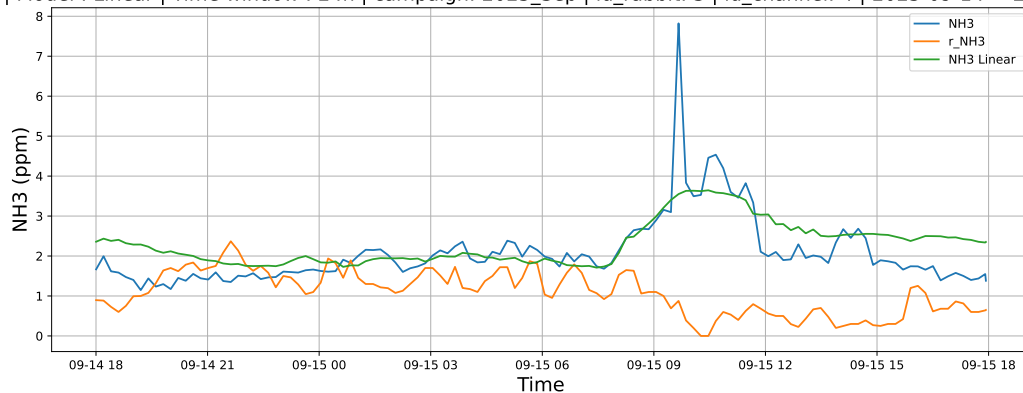
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



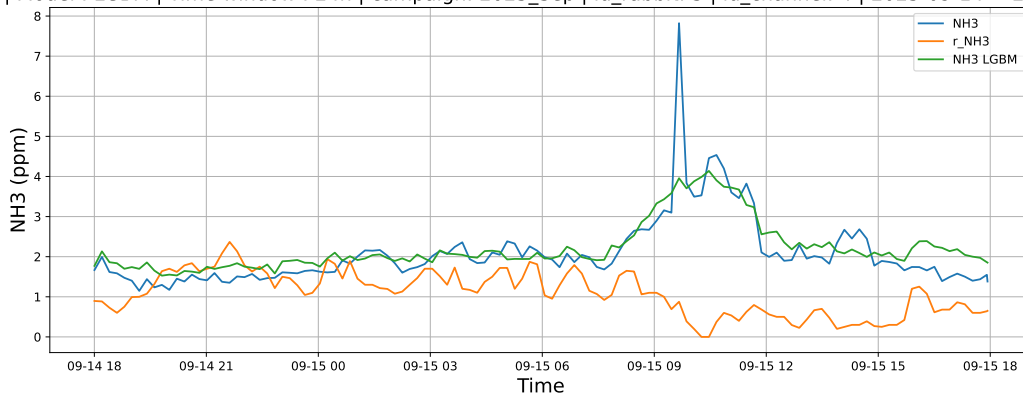
4.1.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

4.1.6.1 Time window : 24

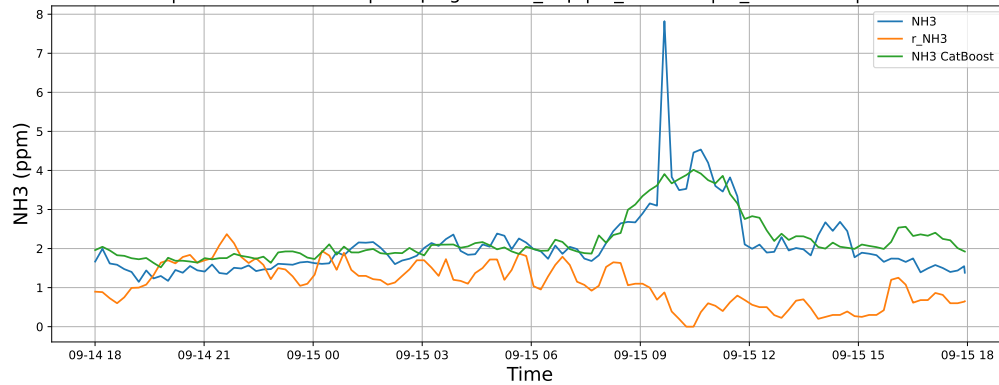
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

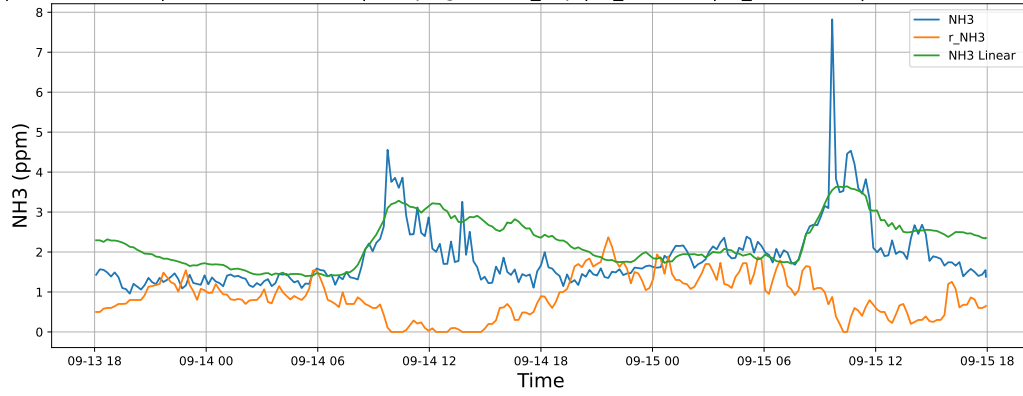


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

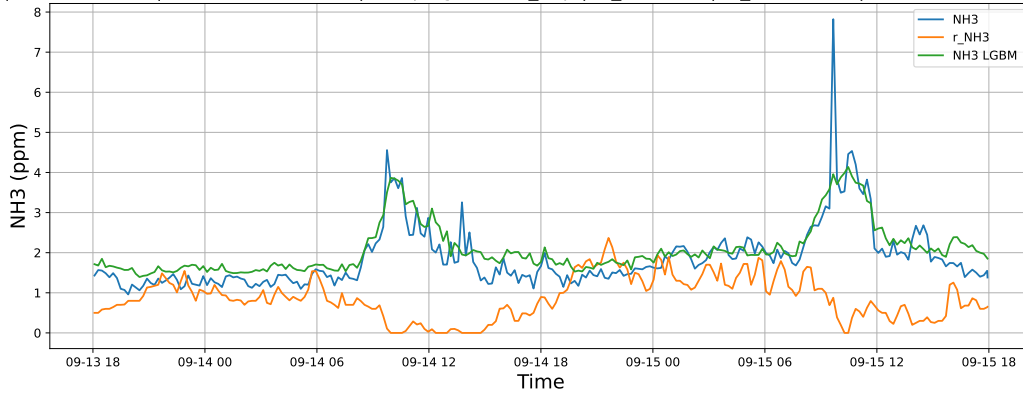


4.1.6.2 Time window : 48

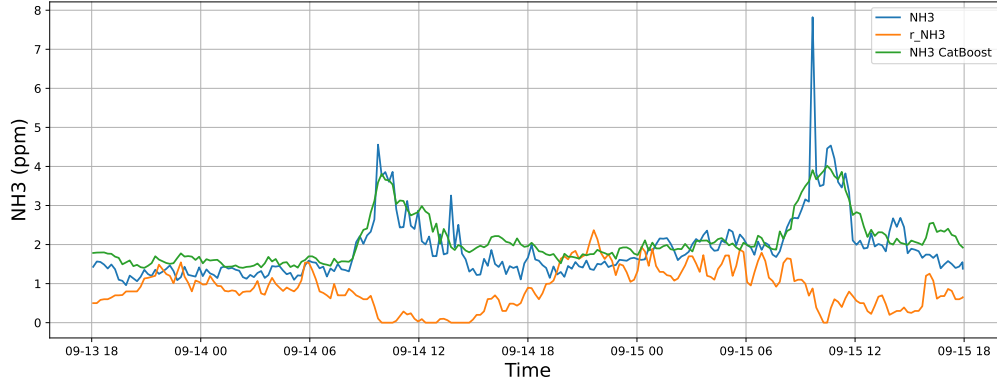
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

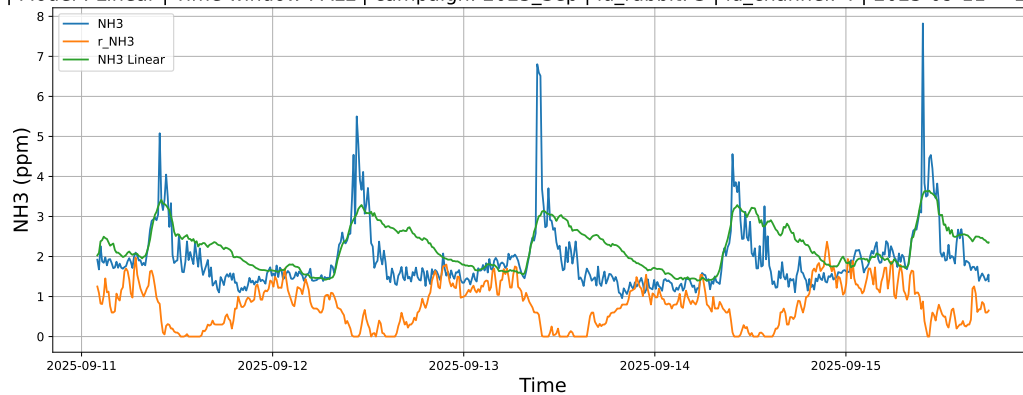


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

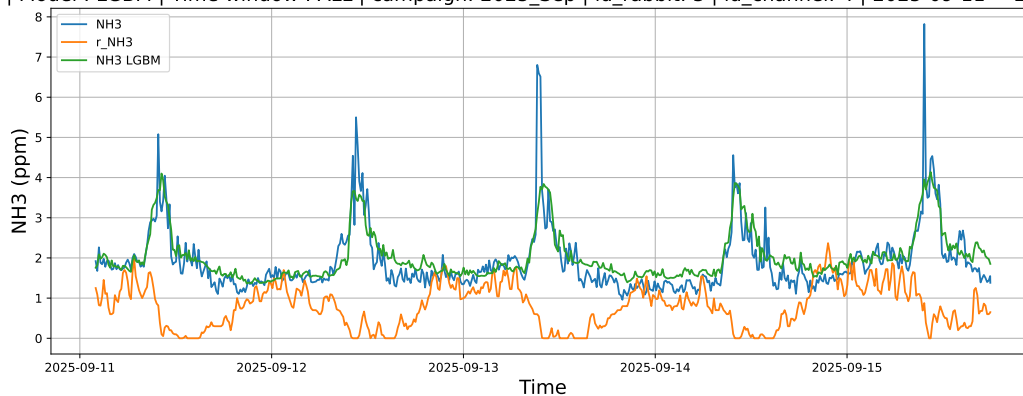


4.1.6.3 Time window : ALL

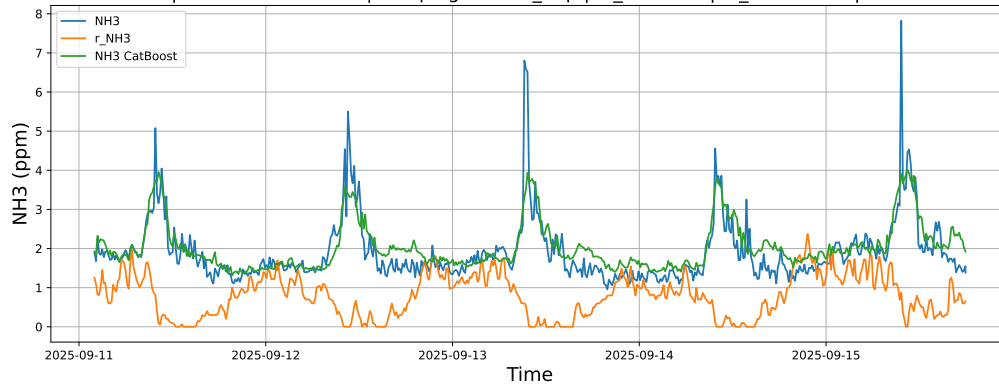
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



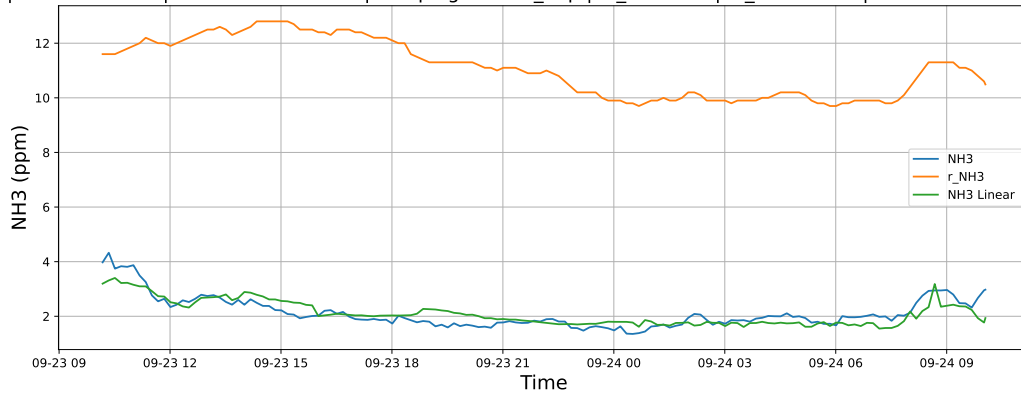
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



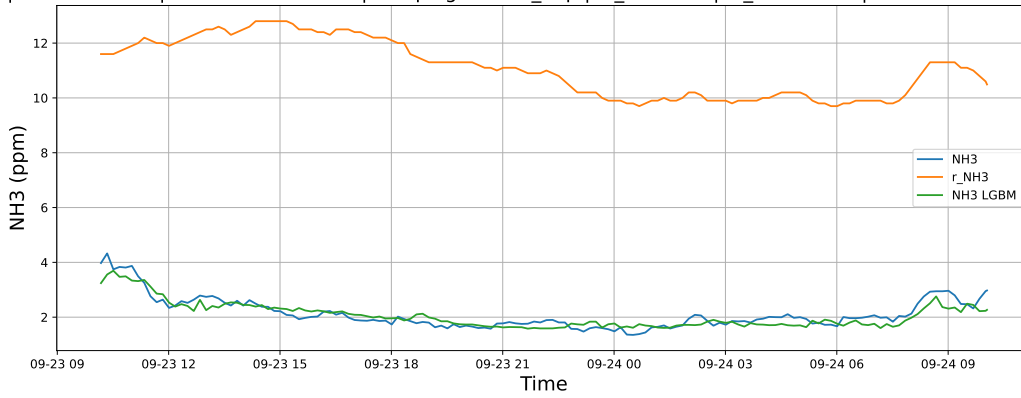
4.1.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

4.1.7.1 Time window : 24

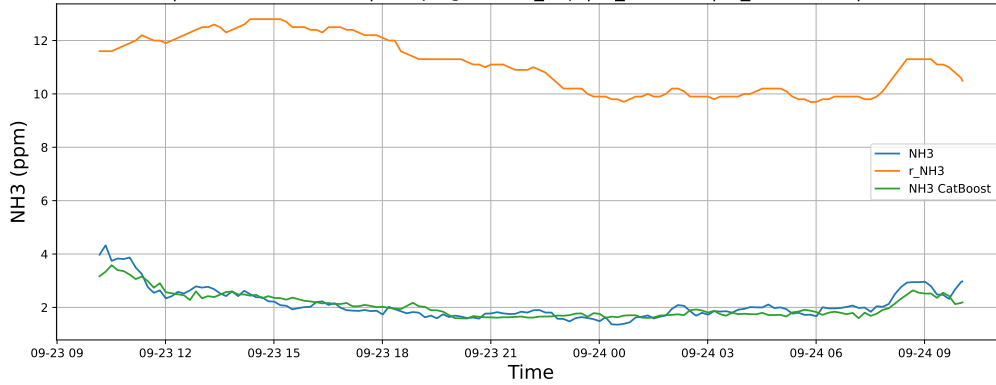
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

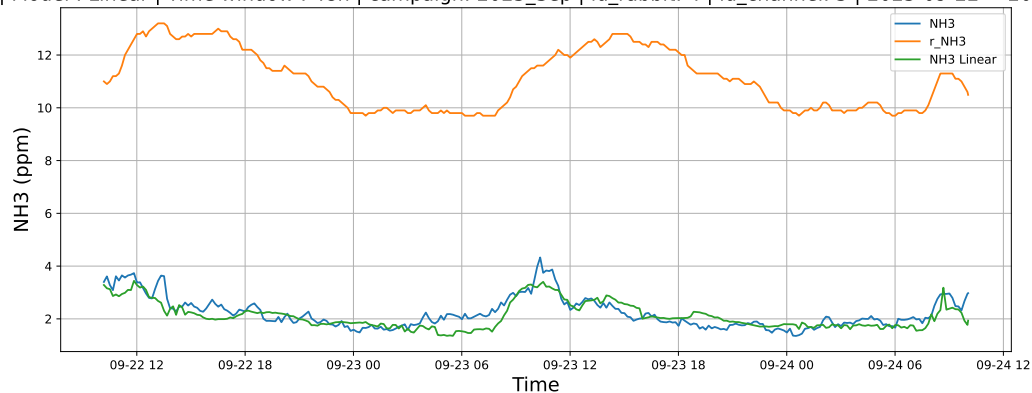


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

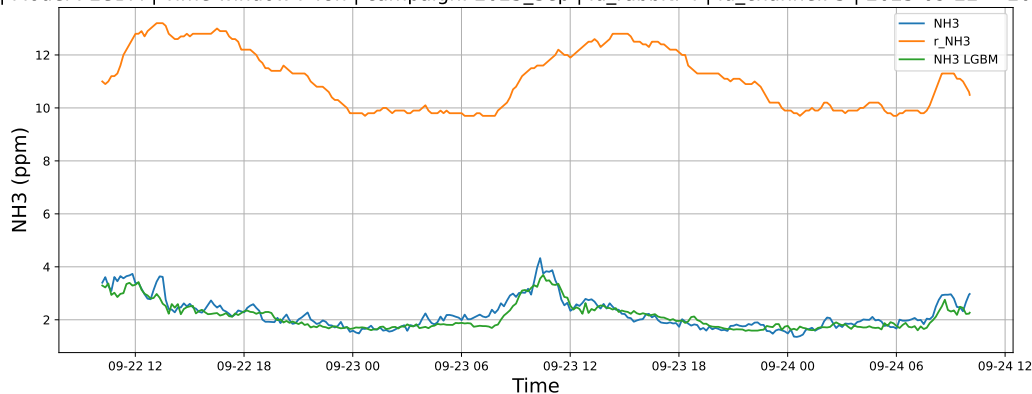


4.1.7.2 Time window : 48

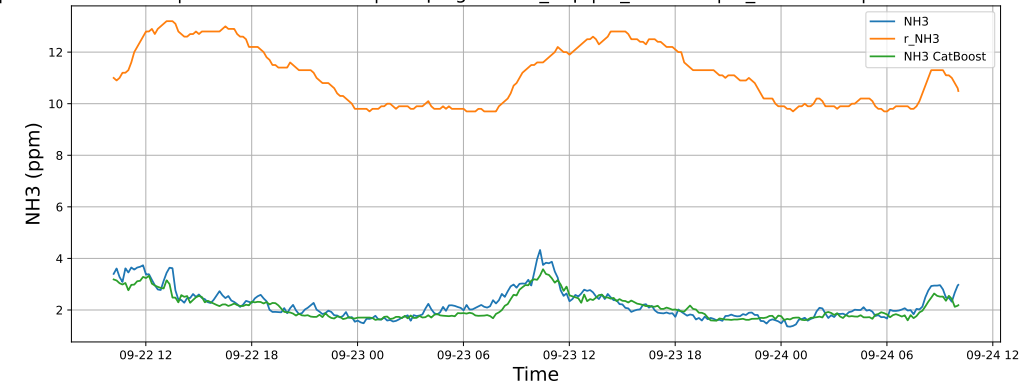
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

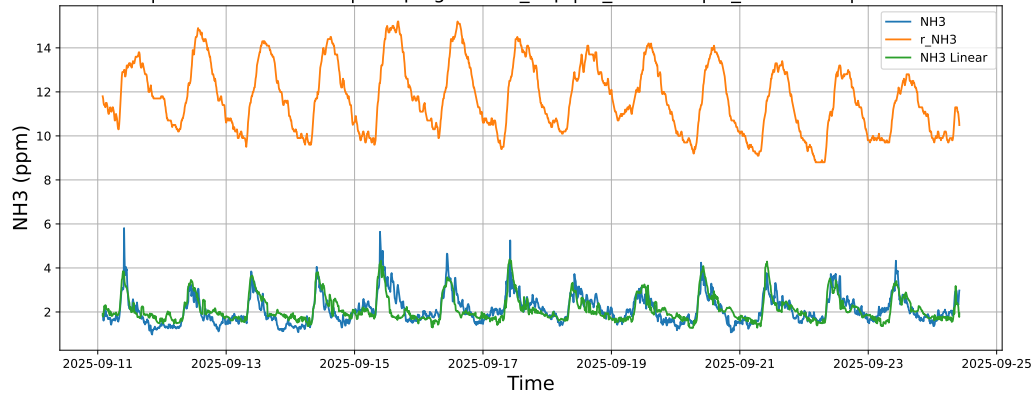


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

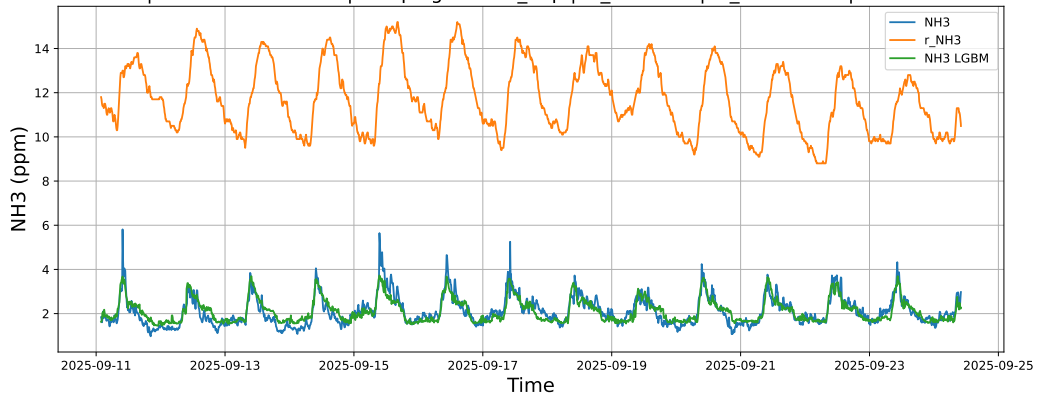


4.1.7.3 Time window : ALL

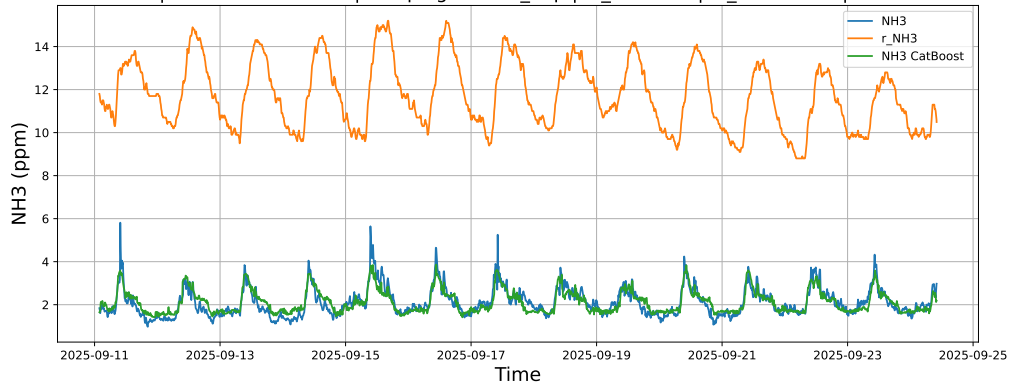
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



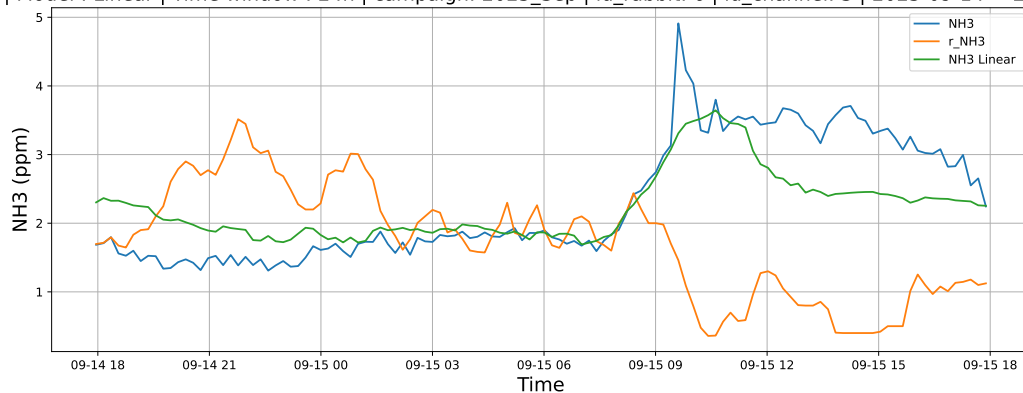
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



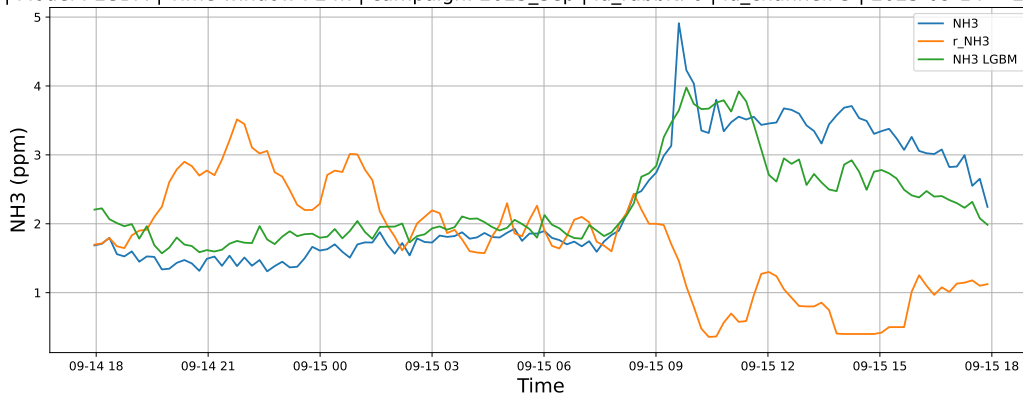
4.1.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

4.1.8.1 Time window : 24

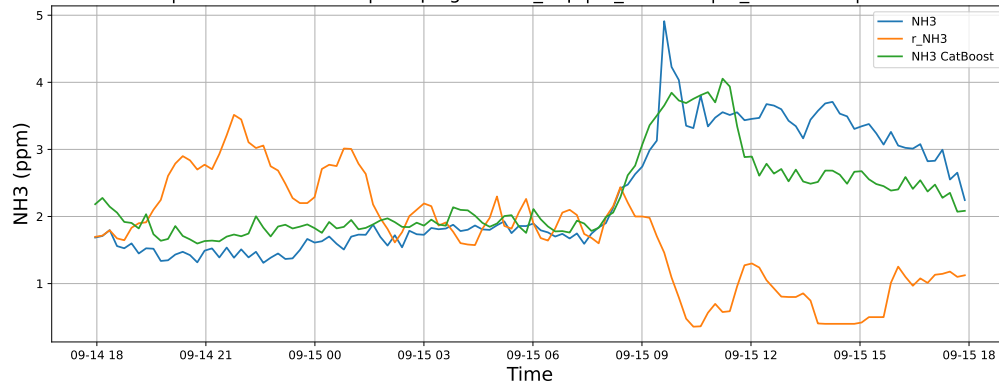
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

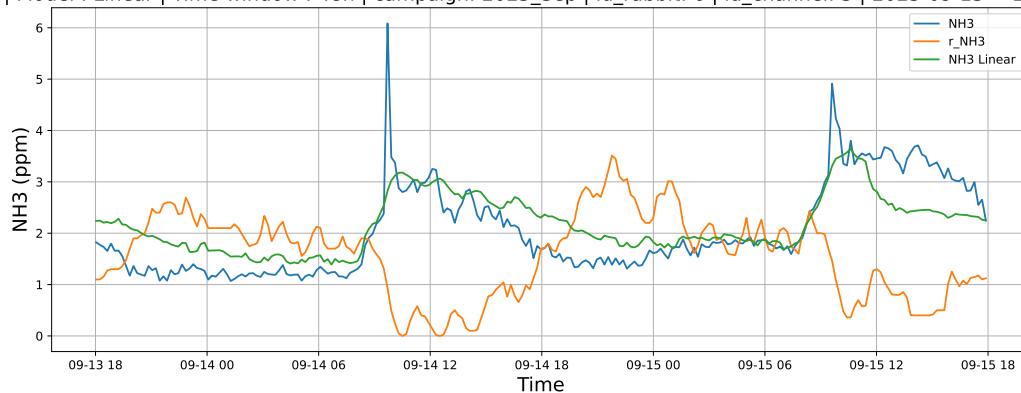


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

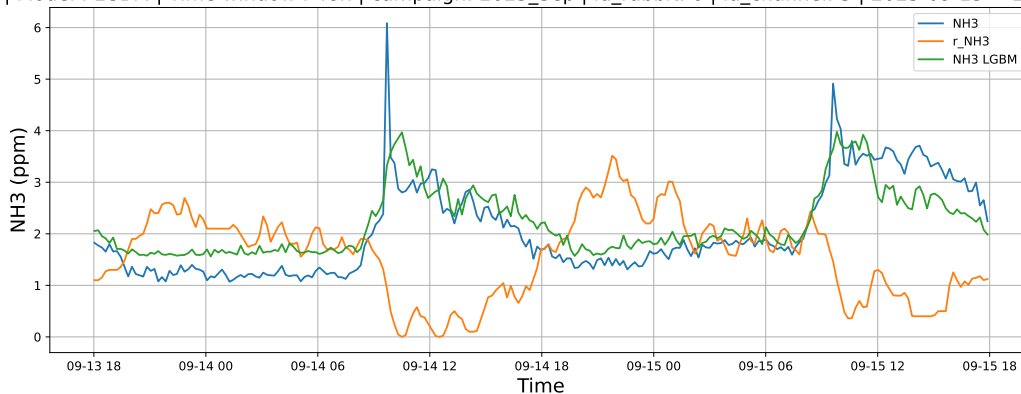


4.1.8.2 Time window : 48

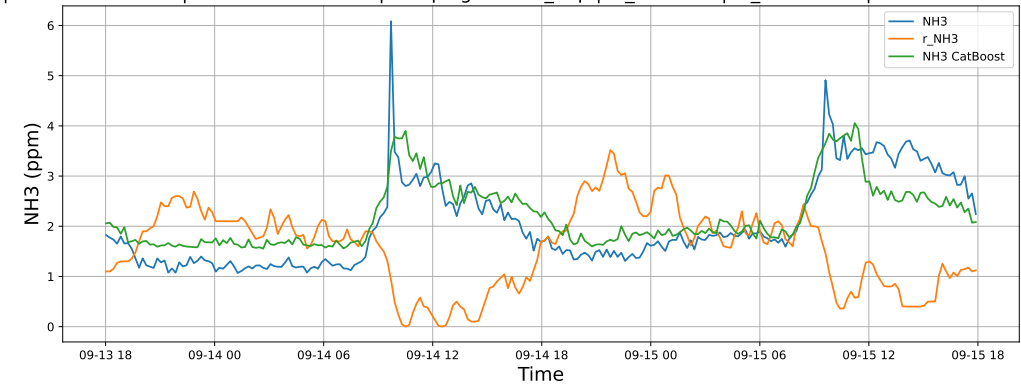
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

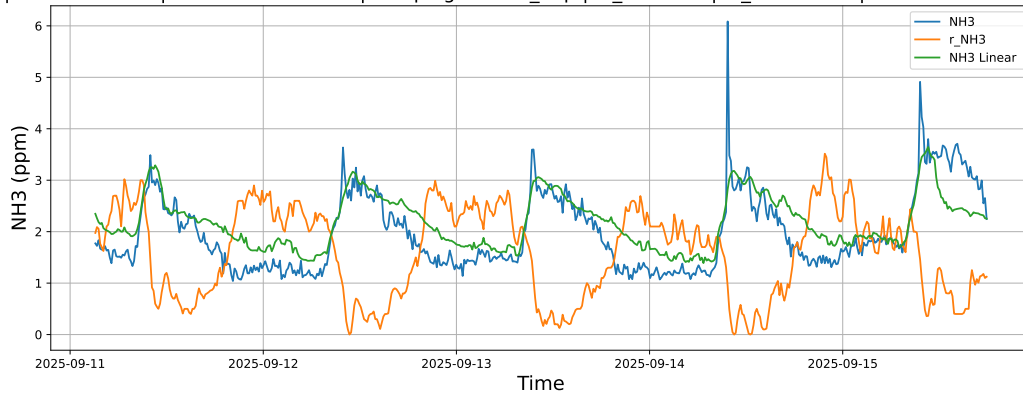


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

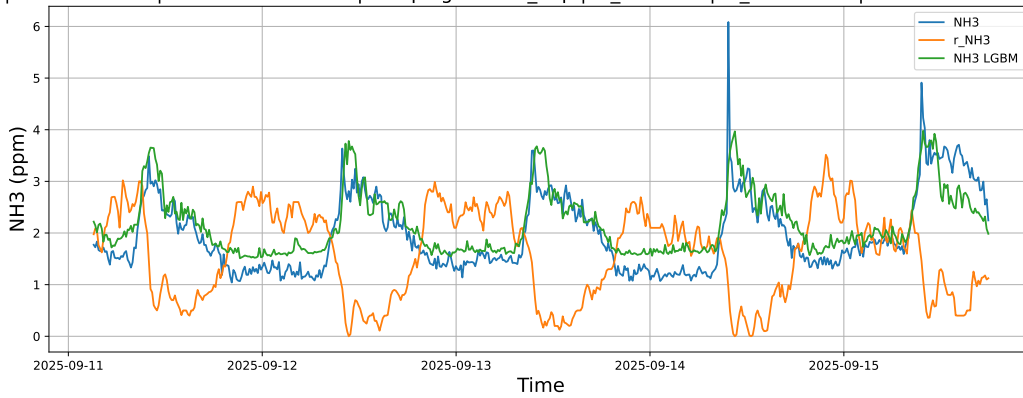


4.1.8.3 Time window : ALL

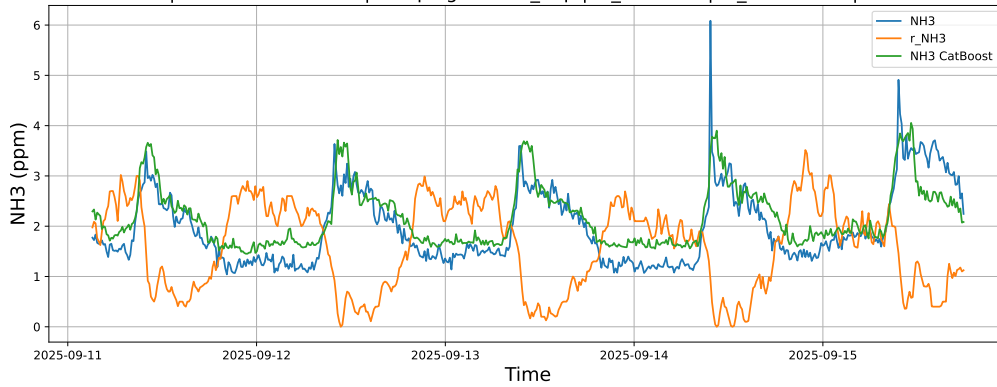
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

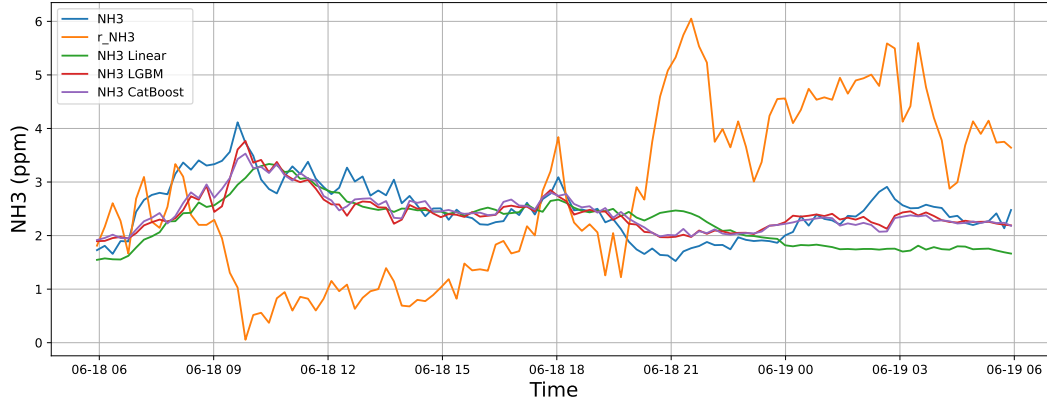


4.2 Compare plots

4.2.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

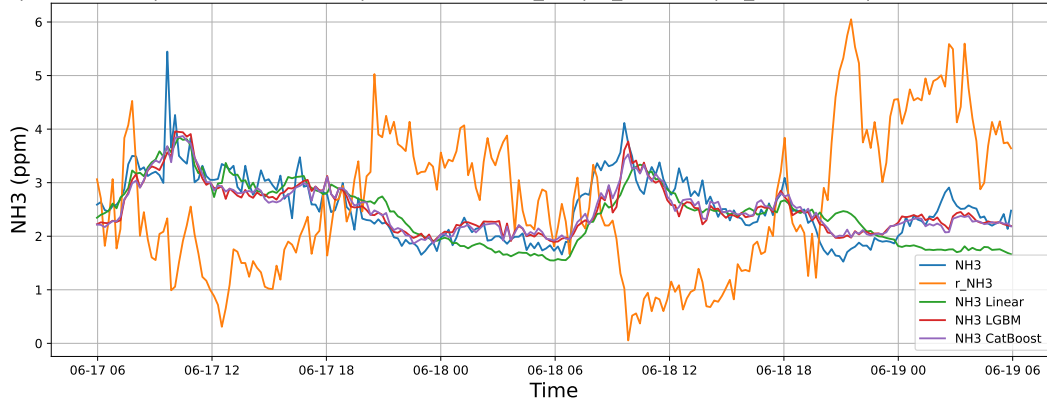
4.2.1.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



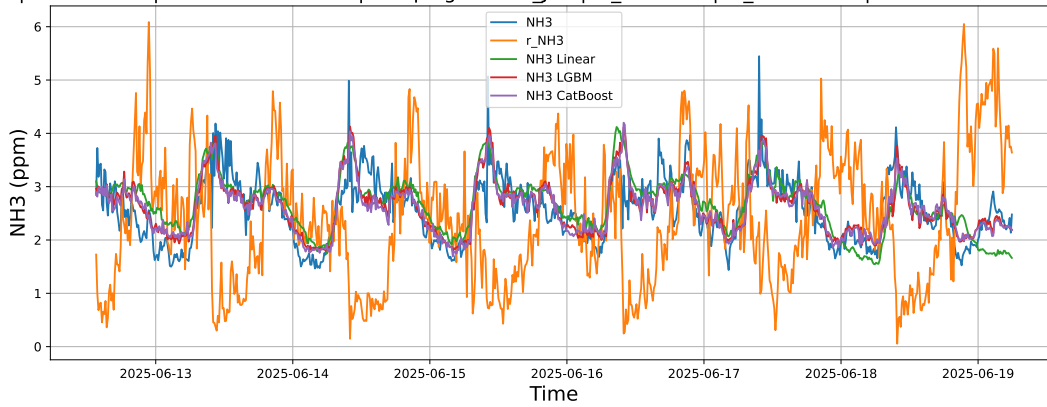
4.2.1.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



4.2.1.3 Time window : ALL

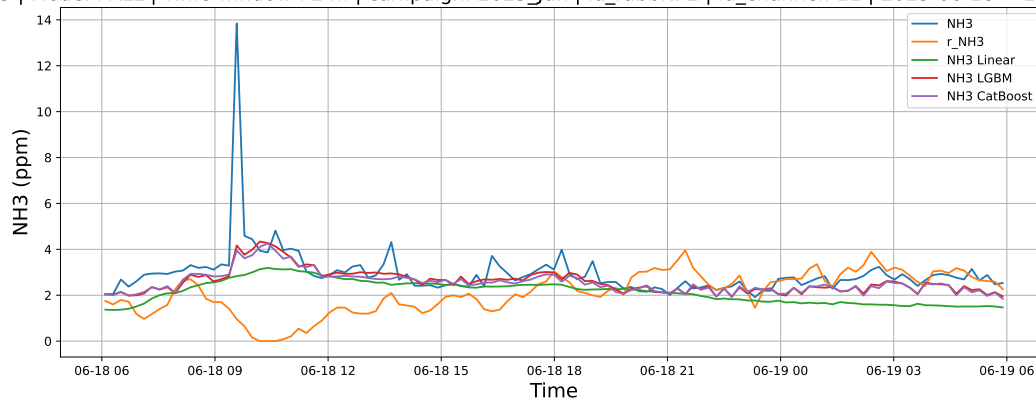
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



4.2.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

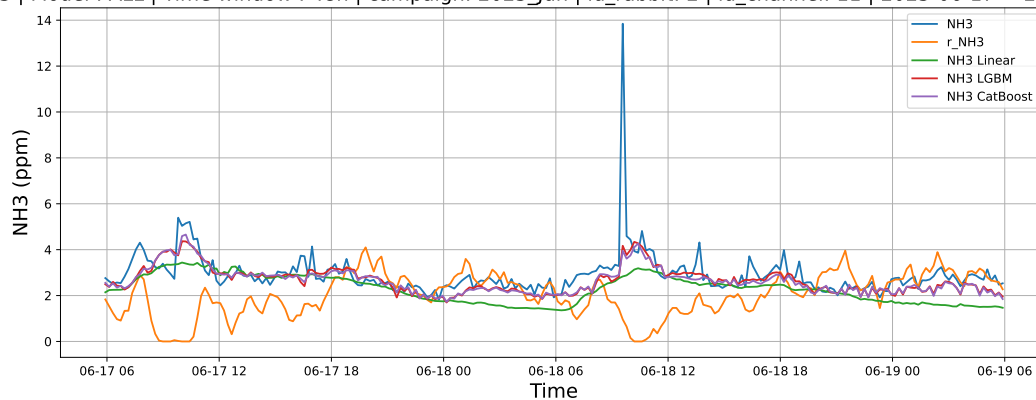
4.2.2.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



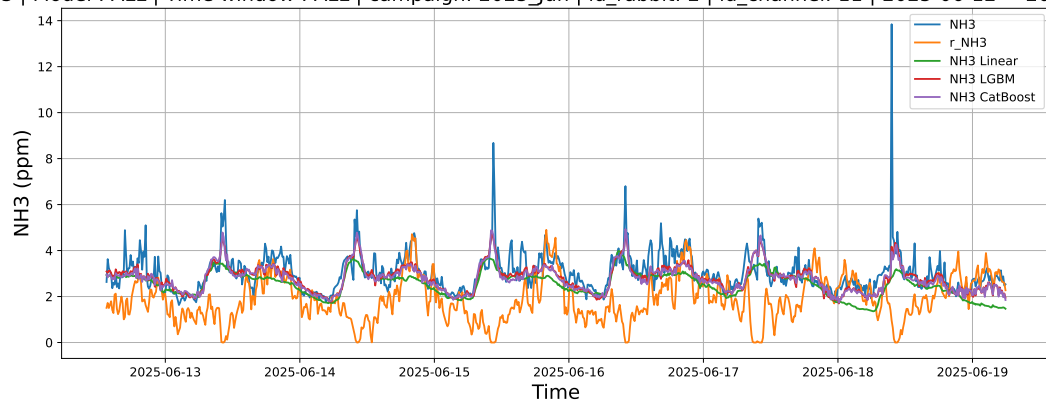
4.2.2.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



4.2.2.3 Time window : ALL

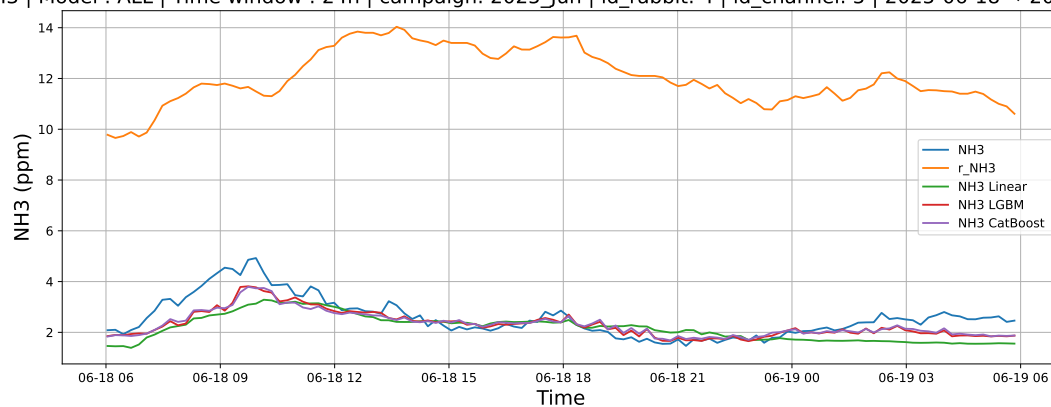
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



4.2.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

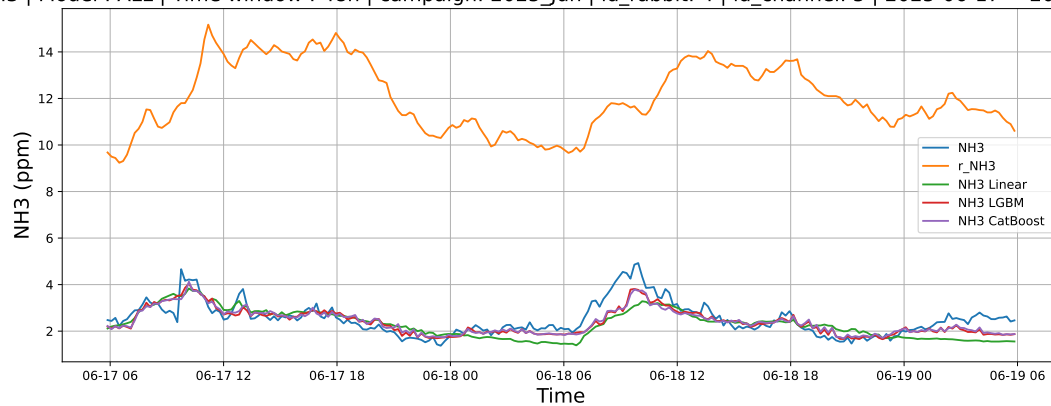
4.2.3.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



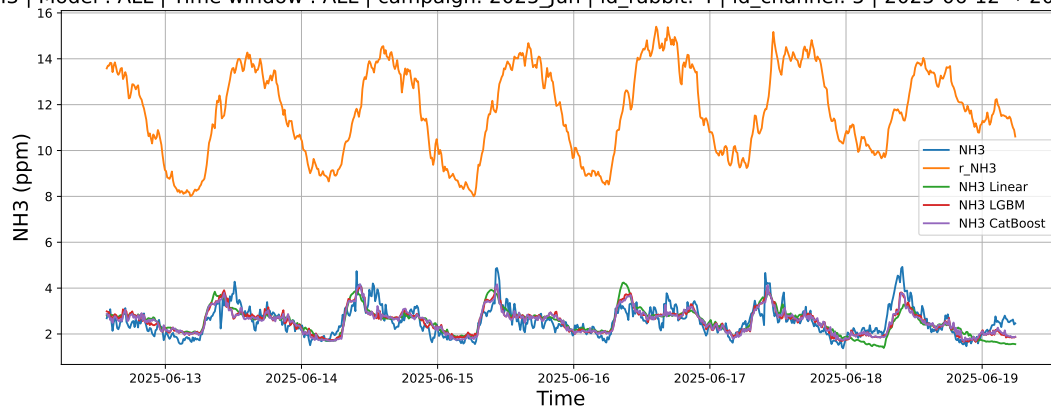
4.2.3.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



4.2.3.3 Time window : ALL

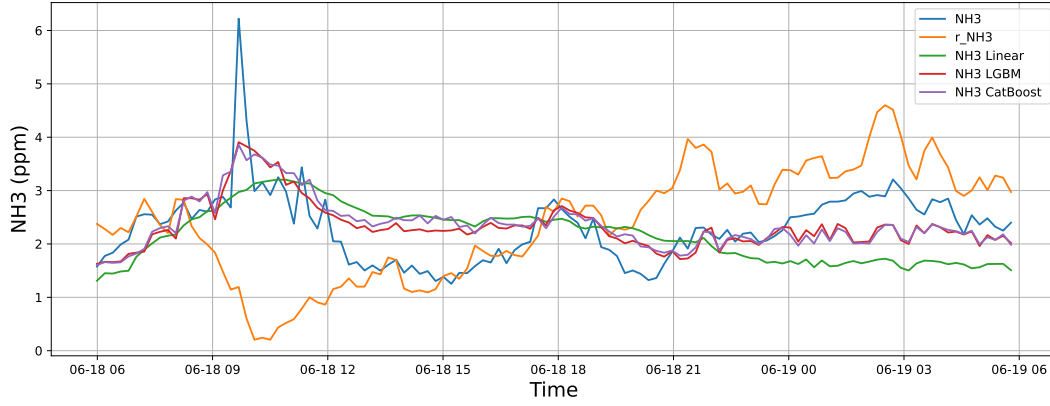
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



4.2.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

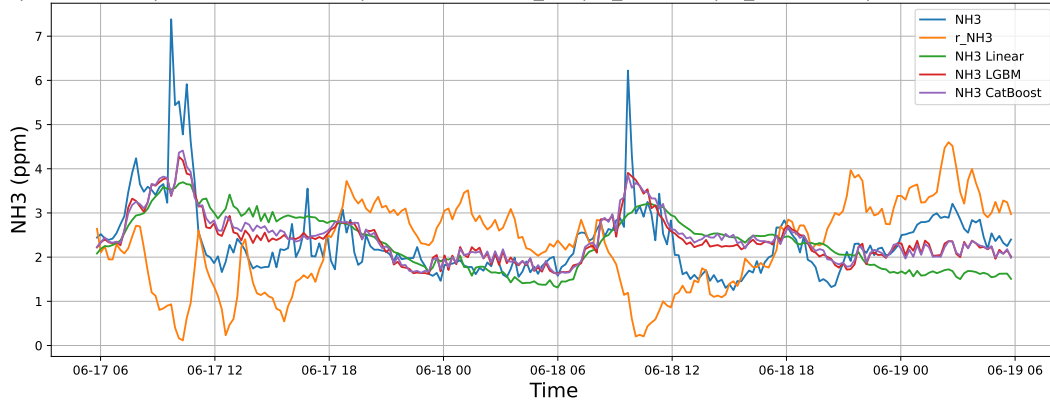
4.2.4.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



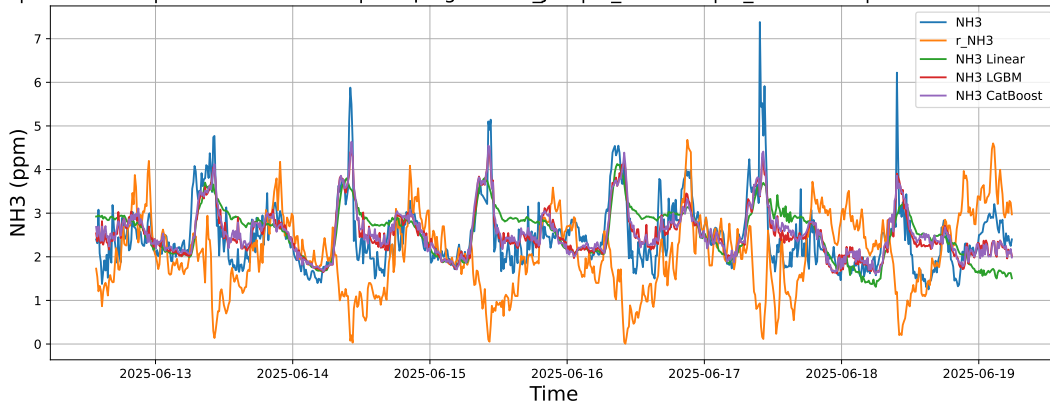
4.2.4.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



4.2.4.3 Time window : ALL

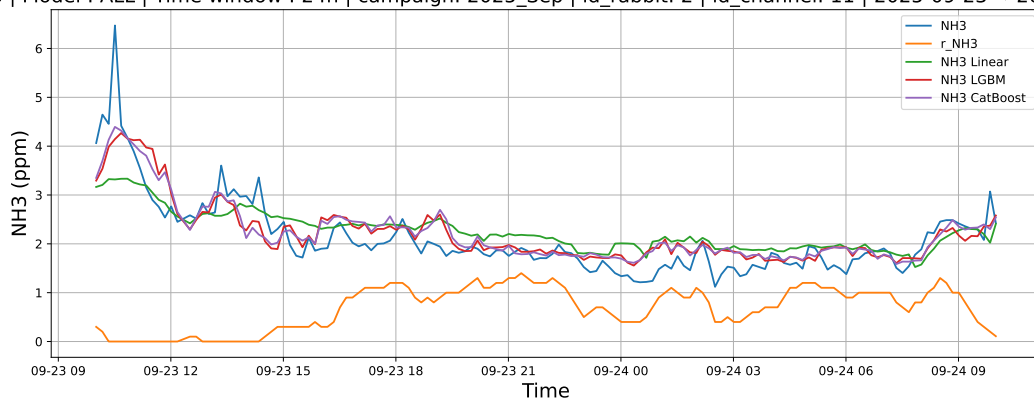
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



4.2.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

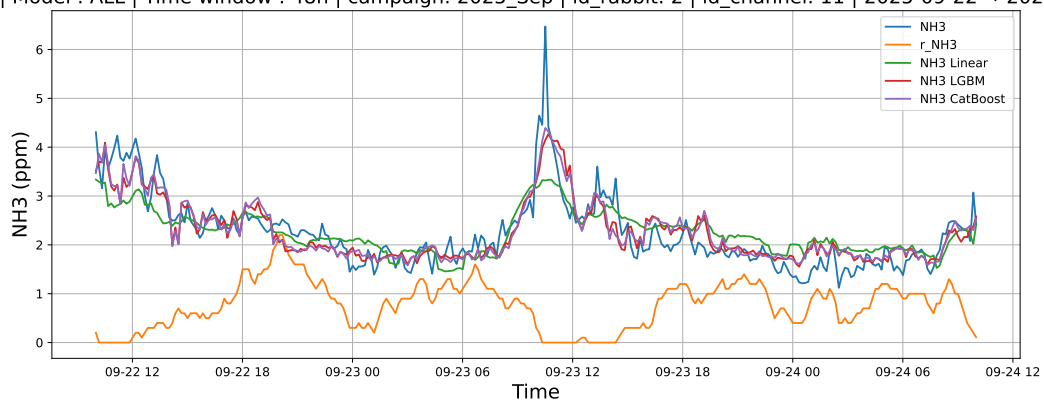
4.2.5.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



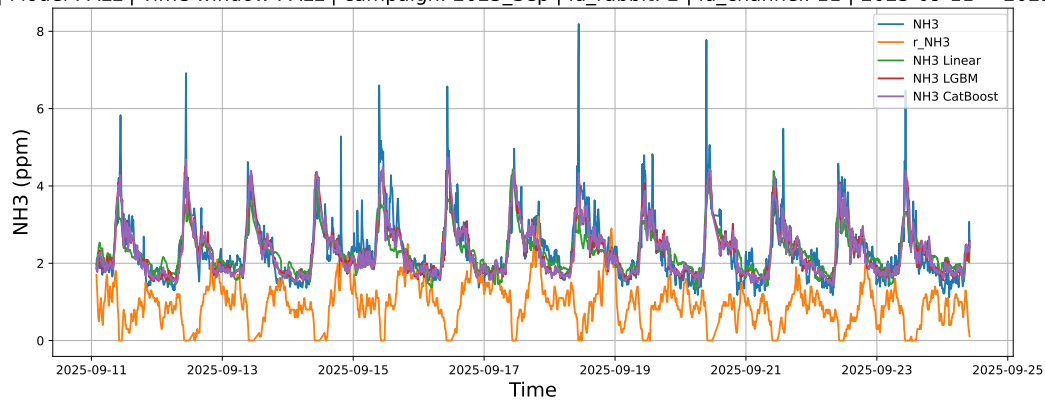
4.2.5.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



4.2.5.3 Time window : ALL

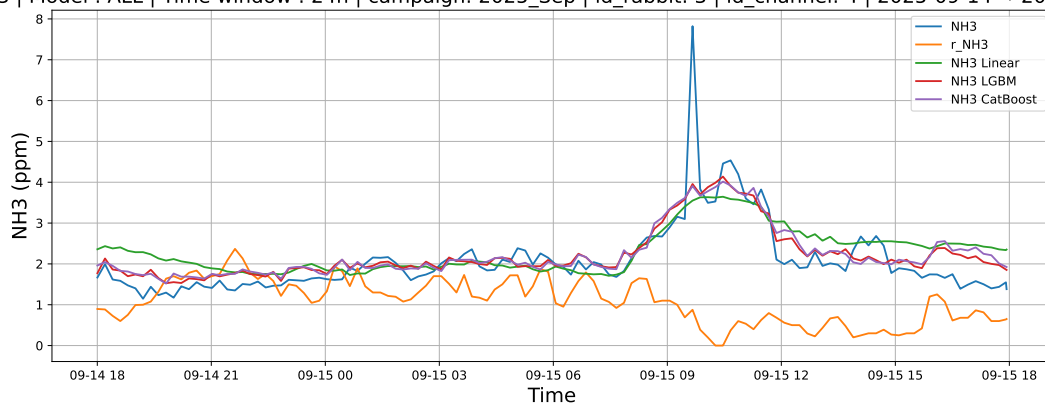
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



4.2.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

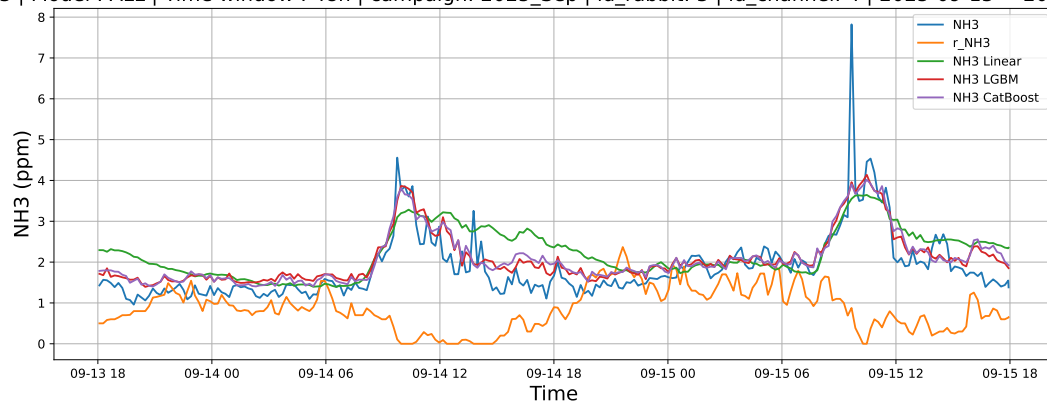
4.2.6.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



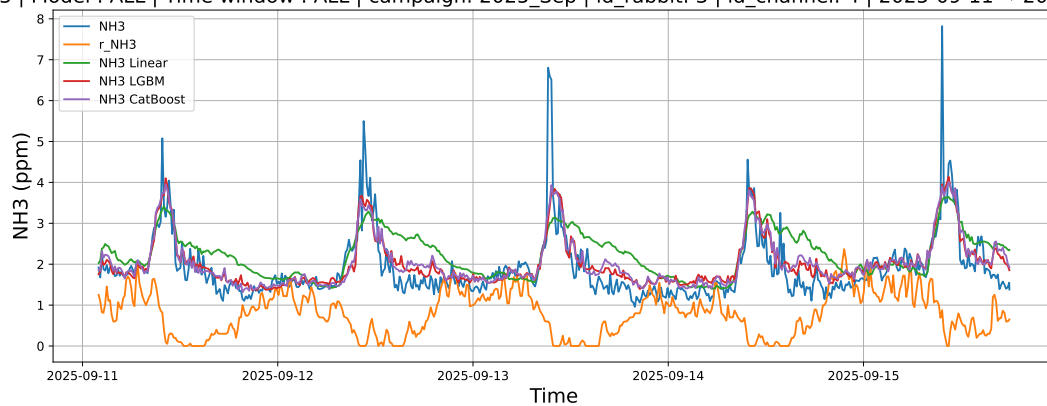
4.2.6.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



4.2.6.3 Time window : ALL

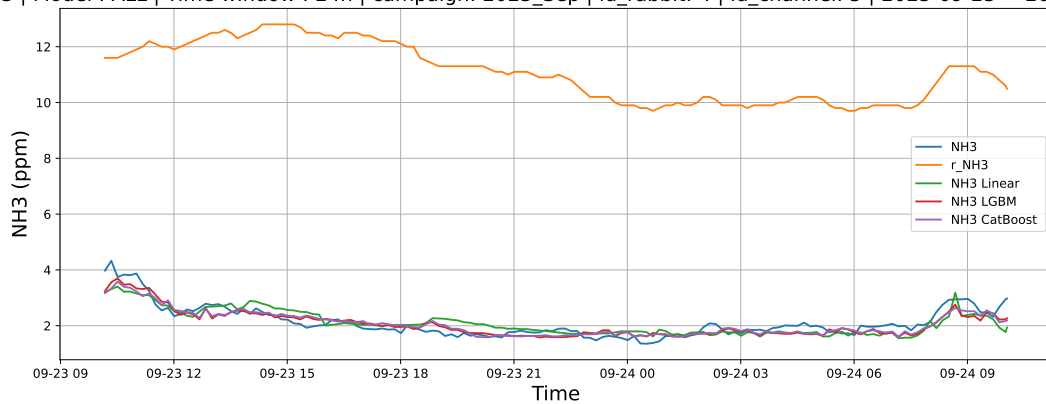
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



4.2.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

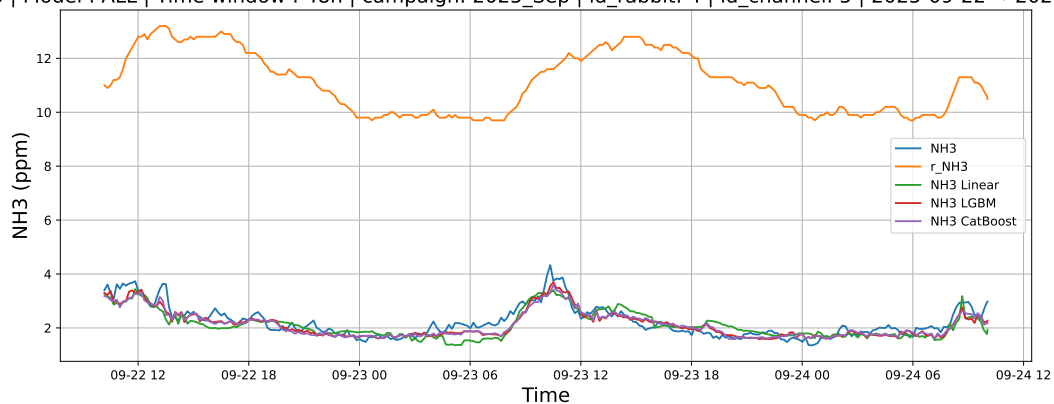
4.2.7.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



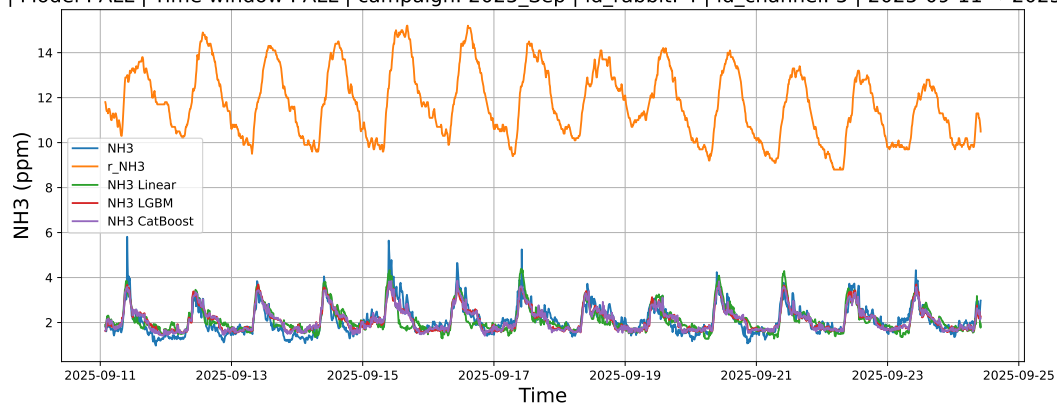
4.2.7.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



4.2.7.3 Time window : ALL

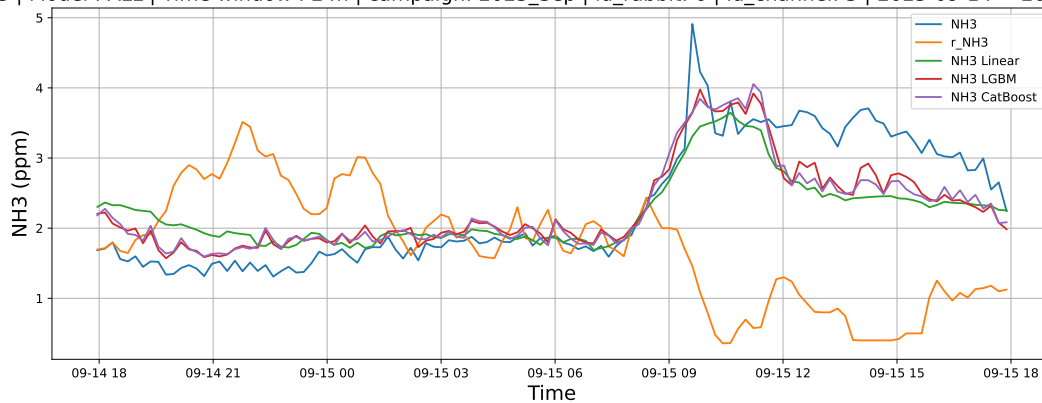
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



4.2.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

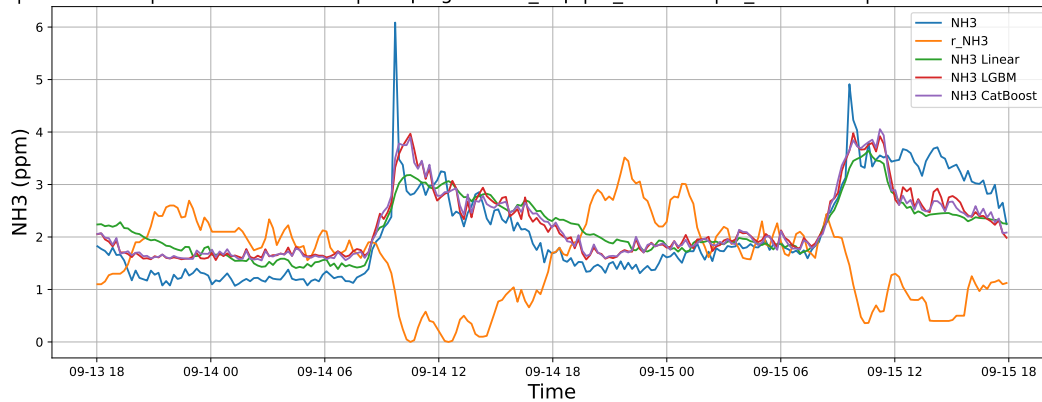
4.2.8.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



4.2.8.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



4.2.8.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

